When Labor Rights Become Actionable: Evidence from Domestic Work in Brazil*

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October 23, 2026‡

Abstract

This paper studies Brazil's Constitutional Amendment 72/2013, which expanded the labor rights available to domestic workers, a largely informal occupation performed inside private homes where government inspections are severely limited. Although the amendment raised the cost of formal employment, many costly provisions were not implemented until 2015. In the interim, the reform generated intense media coverage and public debate about domestic workers' rights. Using an event study design with multiple control groups, we find that the reform increased domestic workers' formal employment rate by 2 to 5 percentage points, raised real wages by 4 to 7 percent, improved minimum wage compliance, and reduced weekly hours and total domestic work employment. The rise in formalization is difficult to reconcile with higher formal labor costs alone. We provide evidence for a worker-side enforcement channel: labor claims involving domestic work increased sharply after the reform, and formalization effects were larger in places where workers had greater exposure to domestic-worker peers and recent formalization events, as well as more pre-reform access to labor courts. The results indicate that labor regulation can increase compliance in hard-to-monitor markets by making legal rights more salient and actionable to workers.

*I thank Ned Augenblick, Leonardo Bursztyn, Stefano DellaVigna, Priscila de Oliveira, Sergio Firpo, Cristine Pinto, Vladimir Ponczek, André Portela, Vitor Possebom, Nick Swanson, Rodrigo Soares, and Gabriel Ulyssea for helpful comments and suggestions. This work was funded by Fundação para a Ciência e a Tecnologia (UID/00124/2025, UID/ PRR/00124/2025, Nova School of Business and Economics DOI: https://doi.org/10.54499/UID/00124/2025), UID/PRR2/00124/2025: DOI: https://doi.org/10.54499/UID/PRR2/00124/2025 and LISBOA2030 (DataLab2030 - LISBOA2030-FEDER-01314200). All errors remain my own.

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‡An earlier version of this paper circulated as Pires (2016), "Labor Rights, Formality and Spillovers: Evidence from Brazil" (available here).

Employers in developing countries frequently evade labor regulations. In Latin America, for instance, nearly 50 percent of workers are employed informally (ILO, 2023). Informality can reduce worker welfare by excluding employees from access to legally mandated benefits, such as maternity leave, unemployment benefits, and social security. Informal employment is also associated with more hazardous working conditions. Beyond the worker's perspective, informality constrains firm productivity by limiting access to credit and weakening incentives for growth, as enforcement of regulations is typically assumed to rise with firm size (Ulyssea, 2018). In addition, a large informal sector shrinks the tax base, leading governments to either cut public goods or raise taxes on the formal sector.

Given the potential consequences, understanding the drivers of labor market informality is critical. In Latin America, contributing factors include low productivity, high labor costs, and limited enforcement of labor regulations (Loayza et al., 2009). Brazil has pursued various strategies to incentivize formalization, including reducing business taxes and auditing workplaces for compliance (Almeida and Carneiro, 2012; Rocha et al., 2018). A less understood margin is whether labor regulation can affect compliance by changing how able workers are to invoke their labor rights.

Domestic work in Brazil includes tasks such as cleaning, cooking, laundry, and childcare. These workers may live with their employers or not. Employment contracts range from full-time to part-time or hourly arrangements. The vast majority of domestic workers are employed informally and they are predominantly Black or mixed-race women from low-income backgrounds. Within this context, Proposal for Constitutional Amendment 66/12 was introduced with the goal of expanding labor rights for domestic workers and elevating their existing protections to constitutional status. This would establish full equality of labor rights between domestic workers and other employees. Importantly, informal employment of domestic workers was already prohibited prior to this reform.

The proposal was approved in April 2013 as Constitutional Amendment 72/13 (CA 72), known as the "PEC das domésticas". It immediately established a maximum of 8 daily

and 44 weekly regular work hours for domestic workers, along with a 50% overtime pay premium. This change increased formal labor costs by around 1%, on average. In June 2015, Complementary Law 150/15 (CL 150) provided the necessary regulatory framework to implement additional rights established under CA 72. These included employer contributions to severance and savings funds, a nighttime pay premium, and access to unemployment insurance, resulting in a more substantial rise in the cost of employing formal domestic workers.

These reforms lacked direct enforcement mechanisms, as the Brazilian constitution prohibits unauthorized government inspections in private residences. As a result, compliance with labor regulations – both before and after CA 72 – relied solely on complaints filed by domestic workers. We document a large increase in labor claims involving domestic work during this period. At the same time, beginning in the period surrounding the amendment, domestic workers' rights received extensive media coverage. Google Trends data show a sharp increase in searches related to domestic workers' labor rights around the passage of CA 72 and the implementation of CL 150. We find a similar pattern by tallying relevant newspaper articles at a leading Brazilian newspaper. We argue that this information shock raised public awareness surrounding the labor rights of domestic workers.

The period between CA 72 and CL 150 – when the law was enacted but key provisions remained unregulated – offers a useful window to assess the effect of the legislation on the domestic labor market through salience of labor rights and worker-side enforcement. This interim period is particularly informative because traditional determinants of formalization decisions, such as government enforcement and labor costs, changed little. We also examine outcomes after CL 150, although the effects of newly implemented labor benefits make interpretation of the results less direct.

We develop a labor demand model of formalization to illustrate the legislation's expected effects on wages and employment. Beyond their effect on mandated labor costs, we model CA 72 and CL 150 as increasing worker enforcement capacity, a margin that may be particularly

important in hard-to-monitor labor markets. This captures the idea that the reforms raised awareness of labor rights and increased workers' perceived ability to file and win claims. Formally, higher enforcement capacity lowers the effective cost of filing a labor claim. As a result, employers face increased expected costs from employing workers informally, while informal workers can bargain for higher pay.

The model yields several key predictions. Higher mandated labor costs reduce formality and employment by raising the cost of formal employment. They also reduce average wages by lowering the share of better-paid formal jobs. In addition, stronger worker enforcement capacity raises formality and wages by increasing the cost of informality, albeit with negative employment effects. Finally, weekly hours may decline if higher labor costs, including overtime pay requirements, reduce demand along the intensive margin.

We examine the labor market effects of the legislation using quarterly individual-level data from Brazil. We use the PNAD Contínua dataset, which has a rotating panel design, from 2012 to 2016. In our main estimation, we apply it as repeated cross sections. We measure formality using two measures: government-registered employment and social security contributions. A time series analysis shows rising formal and part-time employment in domestic work over this period, alongside sizable declines in informal domestic employment, full-time positions, and the total number of domestic workers. These patterns are consistent with our model and suggest that some domestic workers were adversely affected by the reform.

We then analyze the impact of the legislation using an event study framework with multiple control groups, focusing on full-time domestic workers as the main treatment group. To ensure robustness, we consider four control groups formed by combining occupations demographically similar to domestic workers. These groups are formed to approximately satisfy the Stable Unit Treatment Value Assumption (SUTVA). As such, we only include occupations with low transition flows to and from domestic work to minimize potential spillover effects.

We find no evidence of differential pre-trends that would undermine our identification assumptions. Following the implementation of CA 72, domestic workers' formality rates rose by 2 to 5 percentage points over time. This is a noteworthy effect, as higher formal labor costs from the legislation would predict a *decline* in formality. Average weekly hours worked declined by about one hour, while average monthly real wages increased by 4 to 7 percent. The share of workers receiving the minimum wage also rose. These effects are all statistically significant. The reduction in work hours is concentrated among informal domestic workers, while wage gains, being paid the minimum wage and reductions in overtime work are mostly observed among formal domestic workers. Effects generally become stronger after CL 150. One explanation is that full legal codification made these rights more salient and actionable.

Results are robust across specifications, covariate sets, and control groups. In particular, the increases in formality rates and wages persist when using an approximate panel structure, showing the effects are not fully driven by changes in domestic worker composition over time. Results remain similar when part-time domestic workers (*diaristas*) are included in the treatment group. The reforms do not appear to have substantially changed the demographic composition of domestic workers, aside from a modest increase in average age.

We next provide additional evidence consistent with our preferred mechanism: increased worker enforcement capacity. Labor claims involving domestic work rose substantially after the reforms, indicating greater use of legal protections by domestic workers. We then show that the legislation's effects on formalization are larger in neighborhoods where domestic workers have greater exposure to other domestic workers and to recent formalization events. Such interactions may enable workers to compare employment arrangements, learn about legal protections, and update beliefs about the feasibility of pressing for additional labor rights. We also document some suggestive evidence that formalization effects are larger in states with higher pre-reform labor enforcement access, proxied by domestic work legal claim intensity and magistrate density.

We then consider alternative explanations and argue that they cannot fully account for

the results. One narrow interpretation of our mechanism is that the reforms operated exclusively by informing workers and employers about labor rights. We provide some evidence against this interpretation of the results. We also consider alternative enforcement-based explanations. Changes in direct government enforcement are unlikely to be driving the results, since inspections of private households remained legally constrained. The reforms may also have increased the expected legal penalty associated with informal employment by expanding the amount of recoverable benefits and penalties, but this channel is unlikely to be quantitatively large enough to explain the results, since much of the increase involved benefits already owed under formal employment. More broadly, both explanations are difficult to reconcile with the fact that effects are larger in neighborhoods where domestic workers had greater exposure to peers and recent formalization events.

Our paper contributes to several strands of the literature. First, we add to the large body of research that uses quasi-experimental designs to estimate the effects of labor regulation on employment outcomes. This includes studies of minimum wage policies, mandated benefits, and enforcement mechanisms (Almeida and Carneiro, 2012; De Barros and Corseuil, 2004; Cervini-Plá et al., 2014; Samaniego de la Parra and Fernández Bujanda, 2024; Feld, 2022; Kugler, 2004; Lemos, 2004, 2009; MacIsaac and Rama, 1997; Mondino and Montoya, 2004; Theodoro and Scorzafave, 2011).[1] We add to this literature by documenting a case in which expanded labor protections increased compliance despite raising the cost of formal employment. Formalization and wages rose, consistent with the reform making existing and newly expanded rights more enforceable in practice.

Furthermore, our paper is related to research on informal labor markets and the determinants of formality. We draw on the literature on labor market models that incorporate informality and are calibrated to developing country settings (Albrecht et al., 2009; Bosch and Esteban-Pretel, 2012; Haanwinckel, 2025; Haanwinckel and Soares, 2021; Meghir et al., 2015; Ulyssea, 2010). These models typically explain informality through productivity dif-

[1]For broader surveys of this literature in the context of developing economies, see Boeri et al. (2008) and Freeman (2010).

ferences, regulatory costs and limits to state enforcement. We complement this work by showing that worker-side enforceability can also shape the cost of informality. This margin is especially relevant in hard-to-monitor labor markets, where compliance depends on whether workers can turn legal rights into credible claims.

This paper also contributes to work on worker voice, workplace power, and the conditions under which labor protections bind. Recent work studies institutionalized forms of voice, such as worker representation and collective bargaining, and shows that these institutions can affect productivity, workplace amenities, and job quality (Corradini et al., 2025; Harju et al., 2025). We study a setting in which labor protections become effective through workers' capacity to contest noncompliance, rather than through collective or firm-level institutions.

Our work also builds on evidence that information about labor rights can raise wages and improve compliance with mandated benefits (Dinkelman and Ranchhod, 2012; Gindling et al., 2015). We differ from this work by studying a legal reform rather than an explicit informational intervention. The results show that legislation can itself make rights more salient and actionable, linking our findings to work on the expressive effects of law and legal change (Aksoy et al., 2020; Bursztyn and Jensen, 2017; Galbiati et al., 2021; Htun and Jensenius, 2022; Lane et al., 2023; Roeseler and Burns, 2010; Tankard and Paluck, 2017).

We further show that our main effects are larger where domestic workers are more exposed to peers and recent formalization events. This connects our paper to the literature on neighborhood peer effects in labor markets (Bayer et al., 2008; Hellerstein et al., 2014, 2011; Schmutte, 2015). While this literature has emphasized job search, referrals, and employment networks, our setting highlights a different role for local peers: they can amplify the effects of legal change by affecting how workers interpret and respond to new labor entitlements.

The remainder of the paper is organized as follows. Section 2 describes the legislation. Section 3 presents the theoretical model and its predictions. Section 4 outlines the data and empirical strategy. Section 5 reports the results on domestic workers' labor market outcomes. Section 6 investigates the underlying mechanisms. Section 7 concludes by discussing

implications and directions for future research.

2 Context and Legislation

Domestic workers in Brazil perform household services such as cleaning, cooking, laundry, childcare, elderly care, and security, primarily for middle- and upper-income families. Employment arrangements vary, including live-in, full-time, part-time, and hourly contracts. These workers are predominantly Black and mixed-race women from low-income backgrounds. Domestic work remains common in Brazil, reflecting cultural norms, the low cost of low-skilled labor, and conditions rooted in the country's legacy of slavery (Telles, 2014; DIEESE, 2013; Roberts, 2018). These factors contribute to a high informality rate and perceptions of poor working conditions in this sector.

2.1 The Legislation

Constitutional Amendment Proposal 66/12, introduced in December 2012, aimed to modify Article 7 of the Brazilian Constitution to extend the set of labor rights available to domestic workers. Although informal domestic employment was already prohibited by law, the proposal sought to expand labor protections and provide a constitutional basis for existing rights. This would then establish full equality of labor rights between domestic workers and other employees. The Senate approved the amendment on March 19, 2013, and the Chamber of Representatives enacted it on April 2, 2013, as Constitutional Amendment 72/13 (CA 72), commonly known as the "PEC das domésticas."

The political discourse surrounding the amendment emphasized worker dignity, framing the reform as a response to human rights concerns and historical injustices. Political leaders portrayed the exclusion of domestic workers from full labor protections as a legacy of slavery and social inequality. Proponents of the amendment also stressed the need to improve

working conditions through expanded labor protections and mandatory benefits.[2]

Many of the labor rights introduced by Constitutional Amendment 72/13 became enforceable only with the enactment of Complementary Law 150/15 (CL 150) on June 1, 2015, which provided the necessary legal framework for their implementation (Agência Câmara Notícias, 2015; Franco Filho, 2013). Table 1 summarizes the evolution of domestic workers' labor rights. Several key benefits – including the minimum wage, thirteenth salary, paid vacation, four-month maternity leave, 30-day prior notice, and social security contributions – were already established prior to CA 72. Constitutional Amendment 72/13 established a maximum of 8 hours per day and 44 hours per week for regular work, along with a 50% overtime pay premium. Complementary Law 150/15 later introduced additional rights, including nighttime work pay premium, severance pay, unemployment insurance, mandatory employer contributions to the FGTS,[3] childcare assistance, and workplace accident insurance.

Table 2 illustrates the labor cost effects of the legislation through a numerical example. The table reports cost and salary components before and after each legislative change and computes changes in total labor costs relative to the initial scenario. Constitutional Amendment 72/13 introduced components such as overtime premiums, leading to a 3.9% cost increase in this example. More broadly, since approximately 80% of domestic workers were unaffected by overtime provisions, the average increase in labor costs from CA 72 is estimated at roughly 1%, based on pre-Amendment work hours. Complementary Law 150/15 imposed broader mandates – including severance pay and FGTS contributions –

[2]A 2011 article by Senado Notícias reported that then-senator Ana Rita highlighted the vulnerability of female domestic workers, noting that they often suffer "humiliations in the workplace" and are "neglected by legislation itself." According to the senator, "a remnant of slavery" still manifests in the country in the treatment of domestic workers (Senado Notícias, 2011). At the time of CA 72's passage, then-Senate President Renan Calheiros published an article titled A última senzala ("The Last Slave Quarters"), framing the reform as historical rectification. He wrote: "In 1888, the decision to abolish slavery was a decisive step for Brazil to enter the civilized world. By comparison, the exclusion of domestic workers from the set of labor rights was a remnant of that Colonial Brazil marked by backwardness and social injustice." (Calheiros, 2013)

[3]Created in 1967, the Length-of-Service Guarantee Fund (FGTS) serves both as a source of public financing and as a form of insurance against large income shocks. Employers must deposit 8% of each employee's monthly salary into the fund. Workers may access the funds in cases of dismissal, health emergencies or to purchase real estate.

raising employer labor costs by 10.9% relative to the pre-reform scenario in our example.

The amendment included no dedicated enforcement mechanism. Moreover, constitutional protections prohibit unauthorized home inspections by government officials (Franco Filho, 2013; Dolce, 2018). This left enforcement of compliance with labor regulations heavily dependent on worker complaints. If government monitoring is the main mechanism inducing compliance with regulations, this weakens the amendment's practical effect on the labor entitlements received by domestic workers.

2.2 Other Effects of the Legislation

Public and media interest. Despite modest changes in actual labor costs, the period from April 2013 through the passing of Complementary Law 150 saw a surge in media coverage and public attention to domestic workers' rights. Figure 1 displays Google Trends data on related search activity. All six panels show a sharp rise in searches following the legislation, consistent with an information shock that significantly raised public awareness of labor rights for domestic workers. The search terms reflect broad interest in this topic and can be grouped into four categories: (i) rights effective from April 2013 (e.g., "overtime hours domestic worker"); (ii) rights effective from June 2015 (e.g., "FGTS domestic worker"); (iii) formality-related terms (e.g., "legal domestic worker," "rights domestic worker," "law domestic worker"); and (iv) general interest (e.g., "domestic worker").

Figure A1 shows the monthly frequency of newspaper articles on domestic workers published in Estado de São Paulo, a leading Brazilian newspaper, from 2011 to 2016. Results show that media coverage exhibited a sharp spike in March and April 2013 around the approval and enactment of CA 72, and a smaller but clear increase in the second half of 2015, coinciding with the period following the implementation of CL 150. Together, this figure and the last support the view that the reforms sparked a surge in public interest in domestic workers' rights. This provides one reason the reforms could have affected behavior even before all rights were fully implemented.

Labor court claims. We next examine labor-court activity as a measure of worker-side enforcement. In this framework, enforcement need not operate only through government inspections or sanctions. It may also depend on workers' ability to invoke legal protections, either by filing claims or by using the possibility of a claim in bargaining with the employer over wages and working conditions. This channel is especially relevant in domestic work, where, as mentioned before, employment occurs inside private residences and direct inspections are severely limited.

Labor claims are judicial disputes filed in Brazil's labor-court system to resolve conflicts arising from an employment relationship. In the domestic work context, these cases can involve recognition of an employment relationship as well as claims for unpaid wages or benefits.[4] The set of rights that could be invoked in these disputes expanded after CA 72 and CL 150.

Figure 2 reports new labor claims involving domestic work filed in trial-level labor courts (*varas do trabalho*). Panel A plots the annual total number of such claims in Brazil. Claims declined from 31,496 in 2010 to 26,813 in 2013. They then rose to 31,549 in 2014, the first full year after CA 72, and reached 45,994 in 2016, the first full year after CL 150. This represents a 16 percent increase from 2012 to 2014 and a 69 percent increase from 2012 to 2016. Claims remained elevated through 2018. Panel B normalizes claims in each Regional Labor Court (TRT) to their 2012 level and averages the resulting indexes across all TRTs in Brazil. This averaged index rises from 100 in 2012 to 163.6 in 2013 and 213.2 in 2015, reaching 282 in 2016, showing that claims in this measure nearly tripled after the reforms.[5] Both panels show the reforms coincided with a substantial increase in labor-court activity

[4] A contemporaneous newspaper article, based on a survey by the São Paulo Regional Labor Court (TRT-SP), reported that recognition of the employment relationship was the leading cause of labor claims filed by domestic workers. Other leading claims involved termination-related payments and penalties, work-card annotation, moral damages, social security contributions, transportation benefits, overtime, and maternity-related job stability (Scholz, 2013).

[5] The divergence between Panels A and B is driven by TRT-2, the largest regional court. Recorded domestic-services labor claims in TRT-2 fell by roughly 80% in 2013 and never returned to their previous level. Because claims rose in 22 of the 24 TRTs in 2013, this sharp decline may reflect a reporting or classification issue.

involving domestic workers.

3 A Model of Labor Market Formalization

We model the domestic labor market with employers choosing between offering formal employment, informal employment, or no job. Formal employment requires paying the minimum wage and additional taxes. While informal employment avoids these regulatory costs, it exposes employers to potential legal penalties.

Employer payoffs. The employer's payoff from formal employment (π_F) equals match productivity (y) minus employment costs, which include the minimum wage ($\bar{w}$) and costs from taxes and mandated benefits (τ):

$$\pi_F = y - \bar{w}(1+\tau) + \varepsilon_F \tag{1}$$

where ε_F captures unobserved factors affecting the formal employment decision and follows a type-I extreme value distribution. The employer's payoff from informal employment (π_I) is:

$$\pi_I = y - w_I - C_L + \varepsilon_I \tag{2}$$

where w_I is the negotiated informal wage and C_L is the expected legal cost of hiring informally.[6] Both terms are derived below. The term ε_I captures unobserved factors affecting informal employment decisions and follows a type-I extreme value distribution. The employer's payoff from not hiring any domestic worker (π_N) is:

$$\pi_N = b_e + \varepsilon_N \tag{3}$$

where b_e is the employer's outside option and ε_N follows a type-I extreme value distribution.

[6]Formal employment may also raise productivity by increasing workers' effort or attachment to the job, consistent with efficiency-wage mechanisms and improved perceived status at work. We abstract from this channel here.

Employee-initiated enforcement. A worker hired informally can file a labor claim against the employer. Filing yields an expected benefit $pF > 0$, where p is the probability of winning and F is the associated legal financial penalty.[7] Filing a claim also entails an idiosyncratic cost κ, which may reflect time costs, administrative burdens, or other non-pecuniary barriers. We assume κ follows the cumulative distribution function $H(\cdot)$, with $H' > 0$.

We denote by $E \in [0, 1)$ the worker's enforcement capacity. This parameter captures both knowledge of the law and the perceived ability to initiate a claim against the employer. This formulation is related to self-efficacy (Bandura, 1977), defined as individuals' beliefs about their ability to perform actions required to achieve desired outcomes, and perceived behavioral control (Ajzen, 1991), which captures perceived ease or difficulty in undertaking a given behavior.

We model enforcement capacity as reducing the worker-specific cost of filing a labor claim. Specifically, E lowers filing costs so that the effective cost of filing is $(1 - E)\kappa$. We assume this parameter does not affect the probability that a worker prevails conditional on filing. The worker files the claim if and only if the expected benefit exceeds the cost, $pF \geq (1-E)\kappa$. The probability that the employer faces a claim is therefore

$$P(\text{Sued}) = H\left(\frac{pF}{1-E}\right), \qquad \frac{\partial P(\text{Sued})}{\partial E} > 0. \tag{4}$$

The employer's expected legal cost from an informal hire is then

$$C_L = P(\text{Sued})\, pF = H\left(\frac{pF}{1-E}\right) pF. \tag{5}$$

Informal wage determination. We model the informal wage as the outcome of bargaining between the worker and employer. We allow the worker's bargaining weight to rise

[7] For simplicity, we assume that legal claims affect payoffs only through a transfer from the employer to the worker, and abstract from other litigation costs and payments to third parties.

with the employer's expected cost of violating labor laws C_L:

$$w_I = b_w + \theta(C_L)\,(\bar{w} - b_w), \qquad \theta'(\cdot) > 0, \quad \theta \in (0,1). \tag{6}$$

The informal wage thus lies between the worker's outside option b_w and the minimum wage $\bar{w}$, where we assume $\bar{w} > b_w$.[8] The worker's ability to file a claim in labor courts affects informal wage bargaining. A higher worker enforcement capacity E raises the probability that the employer faces a claim $P(\text{Sued})$ and hence increases C_L, shifting the negotiated informal wage toward $\bar{w}$.

3.1 Solving the Model

We now analyze the employer's choice between offering formal employment, informal employment or no job. We assume workers always accept job offers. Under the assumption that the error terms follow an independent and identically distributed type-I extreme value distribution, the model yields a multinomial logit structure (Train, 2009). Substituting the informal wage equation, we derive the following choice probabilities:

$$P(\text{Formal}) = P_F = \frac{e^{y-\bar{w}(1+\tau)}}{e^{y-\bar{w}(1+\tau)} + e^{y-b_w-\theta\left(H\left(\frac{pF}{1-E}\right)pF\right)(\bar{w}-b_w)-H\left(\frac{pF}{1-E}\right)pF} + e^{b_e}} \tag{7}$$

$$P(\text{Informal}) = P_I = \frac{e^{y-b_w-\theta\left(H\left(\frac{pF}{1-E}\right)pF\right)(\bar{w}-b_w)-H\left(\frac{pF}{1-E}\right)pF}}{e^{y-\bar{w}(1+\tau)} + e^{y-b_w-\theta\left(H\left(\frac{pF}{1-E}\right)pF\right)(\bar{w}-b_w)-H\left(\frac{pF}{1-E}\right)pF} + e^{b_e}} \tag{8}$$

$$P(\text{No Offer}) = P_N = \frac{e^{b_e}}{e^{y-\bar{w}(1+\tau)} + e^{y-b_w-\theta\left(H\left(\frac{pF}{1-E}\right)pF\right)(\bar{w}-b_w)-H\left(\frac{pF}{1-E}\right)pF} + e^{b_e}} \tag{9}$$

where $P(\text{Formal})$, $P(\text{Informal})$ and $P(\text{No Offer})$ denote the choice probabilities of formal employment, informal employment, and nonemployment, respectively. Let the formality rate $s_F = \frac{P_F}{P_F+P_I}$ represent the share of formal employment within total employment.

The employer's decision probabilities capture the essential trade-offs in the domestic labor

[8] In practice, informal wage negotiations may also incorporate the worker's valuation of forgone labor benefits. We omit this term for simplicity.

market. Formal employment arises when regulatory costs ($\bar{w}$ and τ) are low relative to the costs of informality. These include the expected legal cost C_L and the wage compensation required in informal jobs, which rises in the worker's bargaining power θ. In addition, match productivity (y) must be high enough to sustain formal wages.

Theorem 1 (Comparative Statics on Formality Rate and Unemployment) *An increase in taxes and mandated benefits (τ) reduces the formality rate ($\frac{\partial s_F}{\partial \tau} < 0$) and raises unemployment ($\frac{\partial P_N}{\partial \tau} > 0$). Greater worker enforcement capacity (E) increases both the formality rate ($\frac{\partial s_F}{\partial E} > 0$) and unemployment ($\frac{\partial P_N}{\partial E} > 0$).*

Proof. *See Appendix B.*

Theorem 2 (Comparative Statics on Average Wages) *Let average wages be given by $\bar{W} = s_F \cdot \bar{w} + (1 - s_F) \cdot w_I$. Then: higher taxes and mandated benefits (τ) reduce average wages ($\frac{\partial \bar{W}}{\partial \tau} < 0$), and greater worker enforcement capacity (E) raises average wages ($\frac{\partial \bar{W}}{\partial E} > 0$).*

Proof. *See Appendix B.*

The intuition is direct. Higher taxes and mandated benefits make formal employment more costly relative to both informal employment and not hiring, reducing the formality rate and increasing the probability that no job is offered. By contrast, greater worker enforcement capacity raises the expected cost of informality. This shifts employment away from informality, increasing the formality rate, but also making some matches unprofitable and therefore raising nonemployment. The wage effects follow a similar logic: higher taxes and benefits reduce average wages by lowering the share of formal jobs, while greater worker enforcement capacity raises average wages by increasing both formalization and the negotiated informal wage.

3.2 Predicted Consequences of the Legislation

The reforms define three relevant regulatory periods: pre-CA 72 (before Q1/2013), post-CA 72 but pre-CL 150 (Q2/2013 to Q1/2015), and post-CL 150 (Q2/2015 onward). CA 72

raised formal employers' overtime costs for work beyond 44 hours per week, leading to modest cost increases. CL 150 increased the cost of formal employment more substantially. We claim both reforms also increased the cost of informal employment. In our framework, this operates through worker-side enforcement: greater awareness of rights raises the probability of labor claims and strengthens workers' bargaining position in informal jobs. This then alters the employer's relative payoff between formal employment, informal employment, and not hiring.

Models of labor formalization typically treat the cost of informality as arising from government fines or inspections. These models would predict a *decline* in formality when formal employment costs rise without a corresponding increase in state enforcement. The formality response is therefore central to our claim that the reform raised the effective cost of employing informal domestic workers even in the absence of an expansion of government monitoring.

The model predicts that higher formal and informal labor costs reduce total domestic worker employment. The period between 2013 and 2015 is especially informative because it allows us to assess the effect of the legislation that operated mainly through workers' increased awareness of labor rights and greater capacity to assert them. Accordingly, the model implies that, prior to the implementation of CL 150, the share of formal domestic employment should rise, as the expected cost of informality increases while there is only a minor uptick in formality costs.

After CL 150, the cost of formal domestic employment rises more significantly, making the net effect on formality theoretically ambiguous. An increase in formality during this period would therefore provide stronger evidence that the reforms increased the cost of informality through worker enforcement capacity. Employers may also exhibit anticipatory behavior before CL 150's implementation, potentially reducing formal employment even before Q2/2015.

We expect a decline in domestic workers' weekly hours after the legislation, reflecting reduced labor demand along the intensive margin. Shorter hours also enable employers to avoid paying for overtime as it becomes more expensive. Per our model, we anticipate a rise in

average wages for domestic workers over this period. In addition to the previously discussed mechanisms, higher labor costs may lead employers to avoid hiring or dismiss lower-skilled workers, shifting the occupation toward relatively higher-paid, more skilled individuals.

4 Data and Empirical Strategy

4.1 Data

Sample and variables. We examine the legislation's effects on domestic workers using the Continuous National Household Sample Survey (PNAD Contínua), a quarterly individual-level dataset in Brazil. The PNAD Contínua follows a rotating panel design, interviewing households for five consecutive quarters. This dataset lacks individual identifiers, preventing us from directly linking observations for the same individuals across time periods. Partially due to this limitation, we primarily analyze the data as repeated cross-sections.[9]

We restrict the sample to individuals aged 15 and above, broadly constituting the working-age population. Employers and self-employed individuals are also excluded from our sample. In our main empirical analysis, we exclude part-time domestic workers (*diaristas*), who are not covered by most labor protections and are therefore less directly exposed to the formal employment margin we study.[10] We incorporate neighborhood-level variables using Primary Sampling Units (UPAs), each comprising at least 60 households and averaging approximately 250 households. UPAs correspond to individual census blocks or aggregates of blocks.[11] The rotating sampling design surveys 14 households per UPA each quarter.

[9]It is possible to match approximately 60 percent of respondents across waves using birth dates, household identifiers, and other personal characteristics. Using this strategy, we later construct an individual-level panel to provide a robustness check of our main results.

[10]We return to this sample restriction below, presenting estimates that include these workers in the treatment group.

[11]"The definition of the UPAs took into account the size of the census tracts; each UPA had to contain at least 60 permanent private households (DPPs), including occupied, occupied without interview, and vacant households. A census tract with 60 or more DPPs constituted a UPA on its own. Census tracts with fewer than the minimum number of DPPs were grouped within the same subdistrict, respecting contiguity, type, and status of the tracts, until the group reached at least 60 DPPs. Each of these formed groups constituted a UPA." (IBGE, 2014)

We use three categories of variables in the analysis: demographics, labor market, and location. Demographic variables include years of schooling, age, race, and gender. Labor market variables include real wages (deflated using the National Consumer Price Index, IPCA), hours worked, formality status, job tenure, and occupation. Formality is measured using two measures – registration with the government in the employee's work card and social security contributions – which are highly correlated ($r = 0.85$). Location variables include urban area indicators, town type, state and UPA identifiers, and UPA-level leave-out means of demographic characteristics.

Descriptive statistics. We use data from 2012 to 2016, ending up with 318,549 domestic worker-quarter observations. The distribution of occupations in our sample of domestic workers is shown in Table A1.[12] Table 3 presents summary statistics comparing domestic workers and other workers across key characteristics. Formal employment is much less common among domestic workers (30 to 35 percent) than other workers (70 to 75 percent). Domestic workers supply fewer weekly hours (33 versus 40) and earn significantly lower monthly wages (R$518.1 versus R$1,172.5). Educational attainment differs significantly, with domestic workers having completed seven years of schooling on average compared to ten years for other workers. In addition, 33% of domestic workers identify as white, versus 44% of other workers. The most pronounced difference appears in gender composition – 91.5% of domestic workers are female, compared to only 36.4% of other workers. Domestic workers are also older, with a mean age of 39 years versus 34 years for all other workers.

Figure A2, Panel (A), shows the distribution of weekly hours worked by formal and informal domestic workers in the year before CA 72. Most work 44 hours or fewer, with formal workers averaging more hours than informal ones. Panel (B) displays the wage distribution. A large share of formal domestic workers earn exactly the minimum wage, while informal ones predominantly earn sub-minimum wages.

[12]General domestic service accounts for 84 percent of the sample, followed by child caretakers (7.5%), home care workers (5.1%), cooks (1.5%), gardeners and horticulturists (0.7%), drivers (0.7%), farmers and skilled gardeners (0.3%), housekeepers and butlers (0.2%), and security guards (0.1%).

Figure 3 plots the evolution of the total number of domestic workers from 2012 to 2016. The vertical red lines mark the implementation of Constitutional Amendment 72/13 and Complementary Law 150/15. Between the pre-CA 72 period and 2016, full-time domestic employment declined by about 5 percent, informal employment fell by 12 percent, and formal employment increased by around 8 percent. We also observe an increase in part-time domestic work (*diaristas*).

Approximately 6.2 million domestic workers were employed in Brazil in 2012 (IBGE, 2019), of whom roughly 78 percent worked full-time and 37 percent of these were formally employed. Applying our estimated percentage changes to the corresponding 2012 stocks, by 2016 full-time domestic employment declined by roughly 240,000 workers. Within domestic work, informal employment fell by about 365,000 and formal employment rose by approximately 145,000. These broad trends indicate a substantial shift toward formalization and part-time work, alongside declines in both informal and full-time domestic employment. The patterns are consistent with the predicted effects of the legislation. They also suggest uneven impacts, with negative consequences for some domestic worker subgroups.

4.2 Empirical Strategy

Event study with a control group. We examine the effects of Constitutional Amendment 72/13 and Complementary Law 150/15 on domestic workers using an event study design with a control group. We set the first quarter of 2013 as the event (t^*), corresponding to when media coverage of the legislation intensified. The estimated equation for outcome y of individual i in occupation o, neighborhood (UPA) s, and quarter t is:

$$y_{iost} = \sum_{k=-4,\neq -1}^{14} \beta_k \mathbb{1}\{t = t^* + k\} \cdot T_o + \beta_E \mathbb{1}\{t > t^* + 14\} \cdot T_o + X'_{iost}\theta + \gamma_s + \delta_o + \eta_t + \varepsilon_{iost} \quad (10)$$

The specification includes fixed effects for occupation (δ_o), quarter (η_t), and neighborhood (UPA, γ_s) to account for unobserved heterogeneity. The treatment indicator T_o equals 1

for domestic workers and 0 for control occupations (detailed below). Standard errors are clustered at the occupation level. The coefficients β_k measure quarter-specific treatment effects relative to the event period t^*, capturing the dynamic impact of the legislation. The omitted category is $k = -1$ (Q4/2012). The coefficient β_E captures long-term effects beginning 15 quarters after t^*.[13]

The covariate matrix X_{iost} includes individual and neighborhood-level controls. Individual controls are third-degree polynomials in years of schooling and age, and binary variables for female and white. Neighborhood (UPA) controls are third-degree polynomials of leave-out means for schooling, age, real wages, and the shares of female and white residents.

Finding a comparison group. To identify the causal effect of the legislation, we require a control group satisfying the Stable Unit Treatment Value Assumption (SUTVA). Occupations with substantial labor flows to and from domestic work should then be excluded to avoid confounding the analysis with potential spillover effects. We construct a variable $trans^o_{DW}$ measuring the share of workers in occupation o at quarter t who were domestic workers in quarter $t-1$. We average this measure over all quarters in 2012 to capture baseline transition patterns. In the resulting distribution, the 25th percentile is 0.1 percent, the median is 0.2 percent, and the 75th percentile is 0.6 percent, indicating limited labor flows between domestic work and most other occupations.

We then build control groups by selecting occupations that are demographically similar to domestic workers and that exhibit low transition rates with them. We calculate average years of education, wages, gender, and racial composition for each occupation in 2012. To avoid hand-picking occupations, we then apply systematic criteria to define four control groups. The race control group includes occupations with $trans^o_{DW}$ at or below the 25th percentile and white worker shares at or below the 75th percentile. The wage control group includes occupations with $trans^o_{DW}$ at or below the 25th percentile and average wages at or below the

[13]Our research design features a binary treatment structure where all domestic workers were simultaneously treated by the legislation, with control occupations being never treated. As such, recent methodological advances in difference-in-differences estimation (Borusyak et al., 2024; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021) are not needed, unlike in settings with staggered treatment adoption.

75th percentile. The schooling control group includes occupations with $trans^{o}_{DW}$ at or below the 25th percentile and average years of education at or below the 75th percentile. The gender control group includes occupations with $trans^{o}_{DW}$ at or below the 50th percentile, female shares above 50 percent, and female individuals only. The four control groups resemble domestic workers in race, wages, schooling, and gender, respectively.

These four distinct control groups provide a robustness check to our estimates of the effects of CA 72 and CL 150 on domestic workers' labor market outcomes. Table A2 reveals that 53.1 percent of workers in our dataset (434,992 individuals) belong to exactly one control group, while 21.9 percent (179,198 individuals) belong to two groups, with the remainder in three or four groups. This limited overlap demonstrates that our control groups are well-differentiated, mitigating concerns that unobserved shocks common to domestic workers and multiple control occupations might drive our results.

The schooling control group serves as the primary comparison group in the main analysis. Appendix Figures present results using alternative control groups, and they confirm the robustness of our findings. Table 4 lists the main occupations in the schooling control group, including secretarial supervisors (5.5 percent), agricultural and industrial machinery mechanics (4.9 percent), machine tool regulators and operators (4.3 percent), and welders and oxy-fuel cutters (4.0 percent). Table A3 reports the corresponding occupations for the other three control groups.

5 Main Results

Labor market outcomes. Figure 4 shows formality rates over time for the treatment and schooling control groups. The vertical red lines mark the implementation of Constitutional Amendment 72/13 and Complementary Law 150/15. Prior to CA 72, both groups exhibit stable formality rates with no differential pre-trends. Following CA 72/13, however, their trajectories diverge. Domestic workers show a sharp increase in formalization, which

continues after CL 150/15. No similar trend is observed in the control group. This pattern suggests that the legislation had an effect on increasing formal employment among domestic workers.

Figure 5 reports event study estimates without controls (Panel A) and with the full set of controls and fixed effects (Panel B). We find the legislation increased domestic workers' formality rates by 2 to 5 percentage points. This effect likely reflects both an increased formalization of existing workers and a reduction in the number of informal domestic workers. This result is consistent with worker-side enforcement and heightened public awareness of labor rights making informal labor contracts more costly to sustain. Otherwise, we would expect a *decline* in formality due to higher labor costs from CA 72 and CL 150.

Figure 6 shows event study estimates for average weekly hours and log real monthly wages. These results, though a little less precise, reveal that weekly hours decreased by approximately one hour, while real wages increased by 4 to 7 percent following the legislation. Figure A3 shows event study estimates for two binary outcomes: working overtime (over 44 hours per week) and receiving the minimum wage. The latter variable can be interpreted as an additional formalization measure. We find some evidence of less overtime work after CA 72. In addition, the share of domestic workers receiving the minimum wage increased by about 3 to 5 percentage points relative to the control group after the legislation. These patterns again suggest that the reforms raised the cost of informality sufficiently to increase compliance even as formal contracts became more expensive.

Figure A4 shows event studies that explore the channels behind our findings by separating domestic workers into formal and informal groups. Several patterns emerge. First, the decline in average hours worked is concentrated among informal domestic workers, suggesting that reduced labor demand – rather than overtime restrictions – drives this effect. This is true despite a fall in the prevalence of overtime work among formal but not informal domestic workers. Both groups see wage gains, but increases are larger for formal domestic workers after CL 150, consistent with selective dismissal of lower-skilled formal workers during this

period. Lastly, the prevalence of being paid the minimum wage rises mainly among formal domestic workers, with only small effects among informal ones.

Our analysis shows that Constitutional Amendment 72/13 and Complementary Law 150/15 affected Brazil's domestic labor market in ways consistent with a higher effective cost of informal employment. Despite raising employer costs, the reforms increased formalization and average wages, while reducing employment and hours worked. The stronger effects after CL 150/2015 are consistent with the law making domestic workers' labor rights more salient and actionable. By fully codifying the rights introduced under CA 72/2013, the law may have increased workers' willingness to contest violations and made noncompliance more costly for employers.

Robustness checks. Our estimates are robust across specifications and covariate combinations. Results remain similar across all four control groups, as shown in Figure A5, Figure A6, and Figure A7. This consistency reduces concerns that our findings are driven by an arbitrary choice of comparison units that violate the parallel trends assumption required for causal identification.

To summarize the impacts of the legislation more concisely, Table A4 presents difference-in-differences estimates of its effects. Formality rates rose by 1 percentage point after CA 72 and by about 5 percentage points (17 percent) after CL 150. Average weekly hours worked fell by about one hour, and overtime incidence declined by 2 percentage points between the two laws. The share of workers paid the minimum wage increased by 2 to 5 percentage points, accompanied by a 4 to 7 percent rise in average real wages. Effects are statistically significant and are generally larger after CL 150.

We replicate our main event study analysis including part-time domestic workers (*diaristas*) in the treatment group. This broader sample allows us to assess whether our findings are sensitive to potential shifts from full-time to part-time domestic work. Figure A8 and Figure A9 report the results for formality, wages, and weekly hours worked. The estimates are generally similar to our baseline findings. Effects on work card formality are smaller – but

become statistically significant over time. This is expected, as most protections introduced by the legislation do not apply to most *diaristas*.

As a robustness check, we replicate the main analysis using an approximate individual panel data instead of repeated cross-sections. Because the data lack unique identifiers, we match individuals across periods using observables such as date of birth and gender, following standard procedures (Osorio, 2022). This matching reduces the sample size and statistical power. The rotating panel structure limits each individual to five quarters of observation. Because each worker is observed for at most five quarters, we combine overlapping panels from successive rotation groups to cover the full analysis period, which then allows us to partly account for time-invariant individual characteristics. Figure A10 shows the results. Estimates for formality rates are consistent with the main findings while being generally noisier. Results for real wages are qualitatively similar, though effect sizes are smaller and less significant. We find no effect on hours worked but observe a large negative effect on the share of workers earning the minimum wage.

Distributional effects. We next examine whether the legislation changed the composition of workers in domestic employment. If the reforms increased the cost of employing marginal workers, the occupation may have become more positively selected, for example with less-educated or otherwise lower-productivity workers exiting domestic work. To assess this possibility, we estimate event study specifications like those in equation (10), but now using workers' demographic characteristics as outcomes and without including control variables.

We report results in Figure A11. Average age increases gradually after the reforms, reaching roughly 0.7 years above the pre-reform baseline, suggesting some exit of younger workers from the sector. The same figure shows no strong evidence of changes in years of schooling. We find some evidence of a decline in the share of women in domestic work, concentrated after the enactment of CL 150, and no strong evidence of a change in the share of white workers. Overall, these patterns do not indicate a large change in the observable

profile of domestic workers following the legislation.

The absence of large changes in worker demographics does not imply that the reforms had no distributional effects. As mentioned before, the time series evidence in Figure 3 shows a shift in the composition of domestic employment following the reform: formal and part-time employment increased, while informal and full-time domestic employment declined. These patterns highlight a tradeoff that may arise when labor standards are strengthened. The legislation improved employment conditions for workers who remained attached to formal domestic jobs, including through higher formalization and wages, but it also reduced overall employment in this sector.

6 Mechanisms

Following Constitutional Amendment 72/13, many informal domestic workers either formalized or exited the labor force. Minimum wage compliance increased after both CA 72 and Complementary Law 150/15. These outcomes contrast with what one might expect following an increase in labor costs that is not accompanied by stronger government enforcement of regulations to induce compliance. This is particularly relevant in our setting, where employment occurs inside private homes and direct state monitoring is inherently limited.

Like in our model, this can be explained by a shift in workers' enforcement capacity. That is, we believe the legislation and associated public discourse increased domestic workers' awareness of legal protections, their willingness to assert labor claims, and their bargaining power in informal employment relationships.[14] Through this channel, the reforms increased the effective cost of employing an informal domestic worker, which matches the empirical patterns presented so far.

This mechanism is consistent with the evidence presented above on public attention and labor claims. Figure 1 and Figure A1 show that the reforms led to a sharp increase in searches

[14]The reform may also have increased lawyers' incentives to represent domestic workers by raising the expected returns to litigation. Greater access to legal assistance could have further strengthened worker-side enforcement, although we cannot separately identify this channel.

and media coverage related to domestic workers' rights, suggesting an information shock that made these rights more salient to both workers and employers. In addition, Figure 2 shows an increase in labor claims involving domestic service after the reforms, indicating that workers became more likely to invoke their labor rights.

We expect enforcement capacity to increase with cumulative exposure to information about domestic workers' rights. Exposure may come from media coverage and public debate, but also from interactions among domestic workers. These interactions can matter because they allow workers to compare experiences, learn whether particular practices are unlawful, and update beliefs about the feasibility of claiming rights. In this sense, the mechanism is related to work on collective workplace power (Corradini et al., 2025): peer communication can help transform worker grievances into rights-based claims.[15]

Motivated by this idea, we test whether the reforms had larger effects where information and discussion about domestic workers' rights were likely more prevalent. Such heterogeneity would support our proposed mechanism and would be less consistent with explanations based solely on formal employment costs or government enforcement. We focus on two measures of formality: government work card registration and social security contributions. The evidence in this section is suggestive.

Number of peers. We now examine domestic workers living in neighborhoods with varying concentrations of domestic workers and formalized workers. The hypothesis is that discussions about domestic workers' labor rights are more common in areas with more of these neighbors. This should then amplify the impact of the legislation on shifting workers' perceived ability to challenge informality in domestic work. Each neighborhood comprises approximately 250 households, with our dataset surveying 5 to 20 percent of residents quarterly. Using this data, we construct proxies for neighborhood-level peer and formalization patterns.

Table A5 shows the relationship between neighborhood composition and domestic worker

[15]Our setting differs, however, because domestic workers are dispersed across households rather than organized within firms.

formalization using pre-CA 72 data. While not causal, the results indicate that domestic workers are more likely to be formal in neighborhoods with higher shares of other domestic workers. Exposure to formal non-domestic workers also predicts a higher chance of formalization.

Next, we implement a triple-differences approach, comparing domestic workers to control occupations before and after CA 72, with heterogeneity by neighborhood peer exposure. The specification includes neighborhood fixed effects and leave-out means of demographic characteristics at the neighborhood level. Table 5 reports the interaction between the difference-in-differences term ($Post \times Treat$) and various measures of the number of peers: domestic workers, non-household domestic workers, recent domestic workers, formal workers, and formal domestic workers.[16] In the post-CA/pre-CL period, an additional neighborhood domestic worker differentially raises formalization probability by 0.7 to 1.2 percentage points for treated workers. Post-CL effects remain positive but are smaller and often statistically insignificant. Estimates using social security contributions as the outcome yield similar effects for neighborhood formal worker density. Overall, the evidence supports the interpretation that peer exposure strengthened incentives for formalization.

Figure A13 and Figure A14 show formalization rates by neighborhood peer density quartiles, using work card registration as the outcome of interest. Results indicate that formalization increased more in neighborhoods with a higher concentration of domestic workers following CA 72. These differences are largest before the passage of CL 150. Results are robust to excluding controls and neighborhood fixed effects.

Formalization events. We now examine formalization *events* rather than the *number* of formal or domestic workers in a neighborhood. For each neighborhood, we compute the average number of formalization transitions over the prior two quarters, separately for

[16]The neighborhood measures have the following means (standard deviations): other domestic workers, 1.16 (1.34); domestic workers from other households, 1.09 (1.28); mean other domestic workers, 1.19 (1.20); formal non-domestic workers, 6.74 (4.21); and other formal domestic workers, 0.37 (0.74). These counts refer to workers observed in the surveyed households within each neighborhood (UPA), rather than the entire neighborhood.

domestic and non-domestic workers.[17] A formalization event occurs when a worker shifts from informal to formal employment status.

Table A6 shows the relationship between neighborhood formalization events and domestic worker formalization using pre-CA 72 data. We find that domestic workers are more likely to formalize when additional other workers in the neighborhood transition to formal employment. All estimates are statistically significant. One interpretation is that peer formalization events facilitate information diffusion and reshape perceptions about informality's acceptability from workers' perspectives.

Table 6 presents triple differences estimates of how legislative effects vary with neighborhood formalization dynamics. Following CA 72, each additional domestic worker formalization event raises a domestic worker's probability of formalizing by 4 to 8 percentage points, depending on the period. Effects are largest and more precise between CA 72 and CL 150. A one standard deviation increase in neighborhood domestic worker transitions increases the treatment effect by 1.2 percentage points during this window. Formalization events among non-domestic workers have positive but statistically insignificant effects.

Figure 7 shows the event study analysis by quartiles of neighborhood formalization events, using work card registration as the outcome. Results are consistent with prior estimates: domestic workers in neighborhoods with more formalization events respond more strongly to the legislation.

Enforcement access. We now study whether the legislation's effects vary with state-level access to labor enforcement. If the reforms operated by raising workers' ability and willingness to contest informal arrangements, their impact may be larger where the institutional means to pursue a claim are more available. We construct two state-level proxies measured before the reform: the intensity of domestic work labor claims normalized by the state labor force in 2012, and the density of first-instance labor court magistrates per 100,000

[17] The mean (standard deviation) number of formalization events is 0.25 (0.40) overall and 0.035 (0.144) for domestic workers. These counts refer to surveyed households within each neighborhood, rather than the entire neighborhood.

inhabitants.[18] Both intend to capture how readily a worker can convert a legal right into an actual claim. We interact each with the treatment indicator in a triple-differences event study. Note that, as these measures vary only at the state level, this exercise is limited in its ability to isolate worker-level access to labor rights' enforcement.

Figure A15 reports the results. We find some evidence that formalization effects are larger in states with higher pre-reform domestic-work claim intensity. The pattern for magistrate density points in the same direction, though it is weaker and less precisely estimated. The direction is consistent with our interpretation: the reforms translated into formalization most where workers had greater practical access to the enforcement channels the legislation activated.

Other mechanisms. The evidence so far raises a further question about the interpretation of our preferred mechanism. One possibility is that the reforms mattered mainly by informing workers and employers about legal rights of which they had previously been unaware. Another is that there was also an increase in workers' perceived ability to invoke those rights. We view the pure information channel as incomplete.

Two pieces of evidence support this. First, as Figure A12 shows, previously formal workers – who were presumably already familiar with their labor rights – experienced similar increases in formalization following CA 72. This is difficult to reconcile with a mechanism based only on informing previously uninformed workers. Second, the broader implication that informality in this sector primarily reflected lack of knowledge of legal rights seems unlikely. While we cannot rule this out, media discussions of domestic workers' labor market in the years before the legislation do not cite lack of awareness as a reason for informality among domestic workers (G1, 2010; Gazeta do Povo, 2010; O Estado de S. Paulo, 2011; Tribuna do Norte, 2011).

[18]The source data are reported at level of Regional Labor Court (TRT), whereas our analysis is conducted at the state level. For São Paulo, which is divided between TRT-2 and TRT-15, we aggregate the two jurisdictions. For the four TRTs that cover two federative units – TRT-8 (Pará and Amapá), TRT-10 (Distrito Federal and Tocantins), TRT-11 (Amazonas and Roraima), and TRT-14 (Rondônia and Acre) – we allocate domestic-work claims across states in proportion to their 2012 populations. For magistrate density, we assign the TRT-level rate to both states in the shared jurisdiction.

An alternative explanation for our findings is that employers expected stricter government enforcement of labor regulations after the legislative changes. While this remains a possibility, we believe it is doubtful this is driving our results. As mentioned before, Brazil's constitution limits home inspections to cases triggered by worker complaints, making broad increases in government-initiated enforcement implausible. Moreover, if enforcement expectations were driving formalization, effects of the legislation would be uniform across domestic workers with respect to the worker's neighborhood composition. Instead, effects are larger for those in neighborhoods with more domestic workers or formalization events.

Another possibility is that employers anticipated the effects of CL 150 following the passage of CA 72. That is, employers might have immediately adjusted their behavior based on expectations about future regulatory changes and cost increases. For one thing, this could confound the identification of the effects of the law in the interim period between the two reforms. However, full anticipation was not possible, as key benefit rates were not agreed upon at the time of CA 72. More broadly, this mechanism cannot account for our results. Rational responses to expected higher costs would predict declining formalization. Yet, we observe gradual increases in formalization rates starting immediately after CA 72.

A remaining alternative is that the reforms increased formalization by raising the expected legal liability associated with informal employment, rather than by changing workers' enforcement capacity. Under this interpretation, the reforms increased the cost of informality through a more mechanical legal penalty channel, closer to a standard enforcement mechanism. This idea is plausible in part, as CA 72 and CL 150 expanded the set of obligations owed by employers to domestic workers. In addition, when these obligations were not paid, employers could be liable not only for the recovery of unpaid benefits, but also sometimes for fines or late-payment charges in labor court disputes. As a result, the amount at stake in some legal disputes increased after the reforms.

We do not view this channel as sufficient to explain our results. First, much of the increase in legal liability corresponds to payments that would have been owed under formal

employment. These liabilities therefore raise the cost of noncompliance, but they do not make formalization relatively less costly than remaining informal. Second, the heterogeneity evidence further points in the direction of our preferred mechanism. Effects are stronger in neighborhoods with greater exposure to domestic worker peers and formalization events. This pattern is more naturally explained by changes in worker enforcement capacity, operating through information diffusion and greater willingness to press employers for labor rights.

7 Conclusion

This paper examines the impact of labor legislation on Brazil's domestic labor market. For two years after its passage, Constitutional Amendment 72/2013 had limited effects on the costs of employing a formal domestic worker as many of its provisions required additional regulation. At the same time, it generated substantial media coverage and public attention to domestic workers' rights. This interim period – during which the amendment had been enacted but most provisions were not yet in place – provides a setting to assess effects other than the increases in taxes and mandated benefits. We also analyze the subsequent effects of the full implementation of these constitutional provisions through Complementary Law 150/2015.

Our findings indicate that this legislation increased formalization through mechanisms beyond traditional policy channels. Despite limited direct enforcement capacity and higher formal employment costs, domestic worker formalization rates rose significantly, while weekly hours decreased and real wages increased. At the same time, total domestic work employment declined. We interpret these effects as evidence that the reforms increased workers' awareness of their rights, strengthened their ability and willingness to initiate labor claims, and raised employers' expected cost of maintaining informal contracts in a market that the state cannot directly monitor.

The legislation's impact also varied systematically across settings where these mechanisms should be more powerful. Domestic workers in neighborhoods with higher concentrations of other domestic workers, those exposed to more recent domestic-worker formalization events, and those in states with greater pre-reform access to labor enforcement experienced generally larger increases in formalization. These patterns suggest that information about rights, formalization, and the feasibility of claims spread through local networks, consistent with our preferred mechanism.

Our results suggest that labor regulation can increase compliance even when direct enforcement is weak. This mechanism is relevant beyond domestic work, including in small firms and some segments of agricultural and service employment where violations may be difficult to monitor directly. The findings point to legislation that raises awareness of rights as a complement to state enforcement in hard-to-monitor labor markets.

These results do not imply that formalization is always welfare improving. In general, informal employment may be less costly for some higher-wage workers, who may value flexibility or higher take-home pay more than formal labor protections. The low-skill nature of the occupation we consider in this paper likely renders the trade-off between formal and informal contracts more clear-cut. Nevertheless, even in this context, formalization may not be universally preferred by workers. The employment effects also matter: a higher expected cost of informality may increase formalization while making some low-productivity matches unprofitable. A formal welfare analysis that accounts for these competing effects is left for future work.

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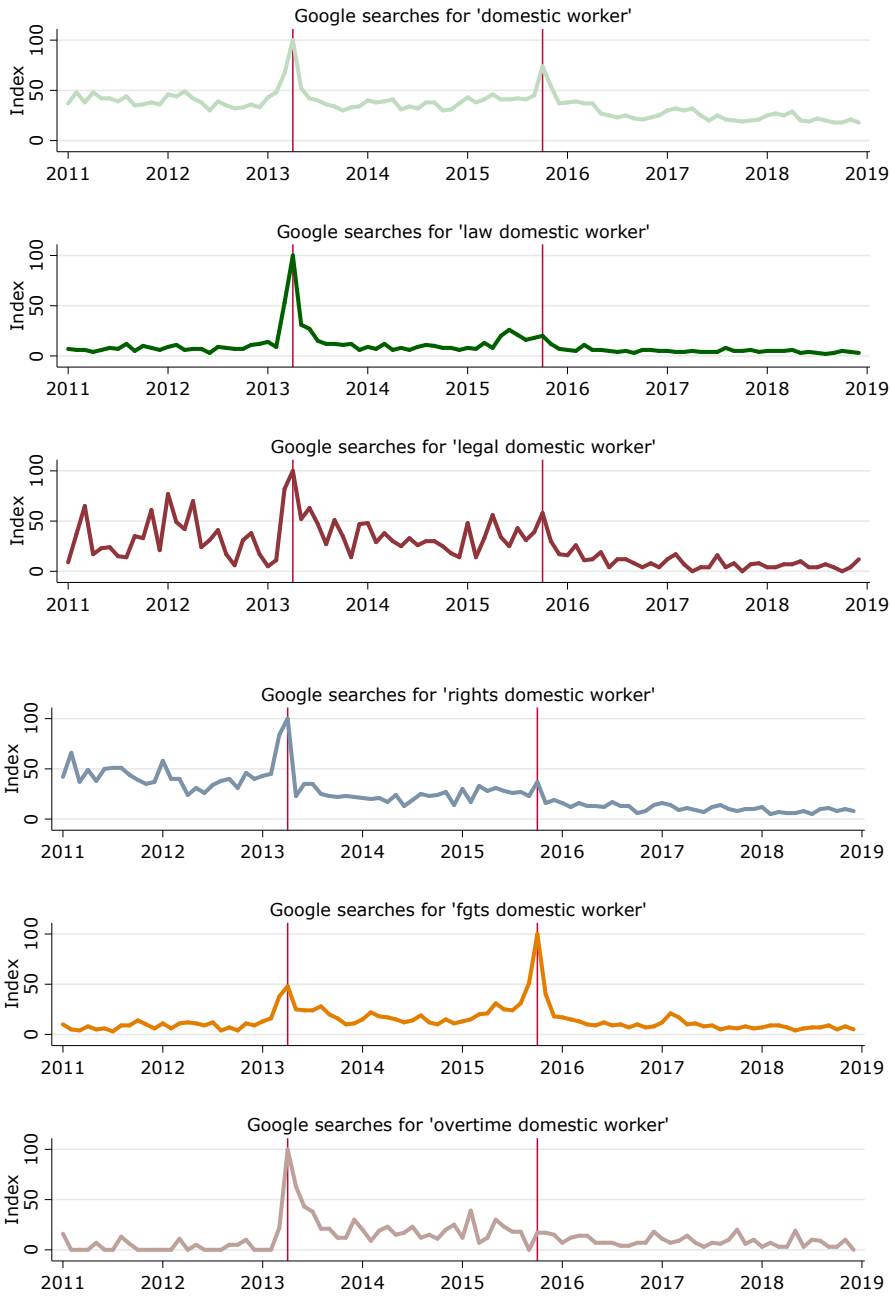
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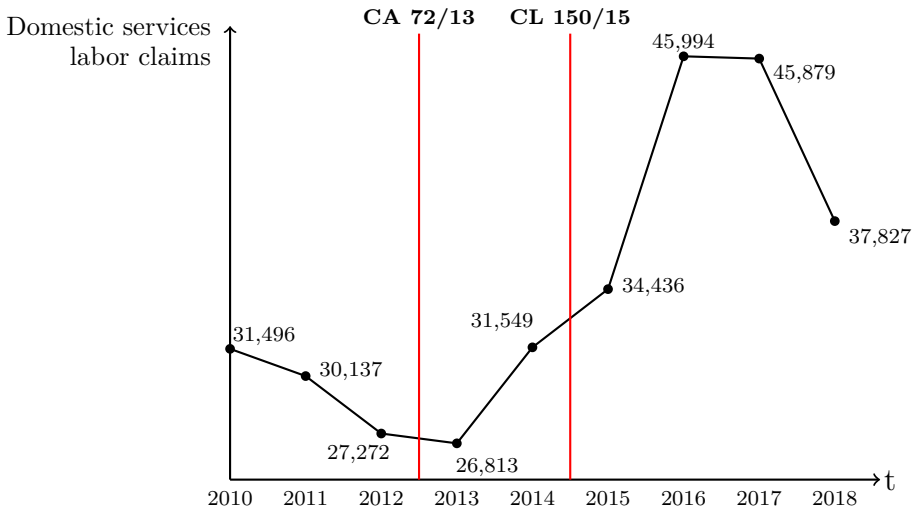
Figures

Figure 1: Interest on topics related to domestic workers around CA 72 and CL 150

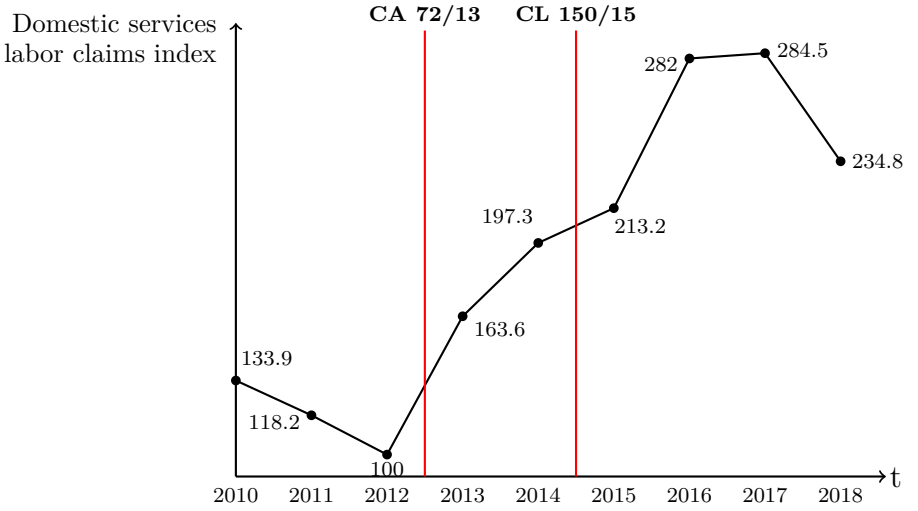


Notes: Weekly data from Google Trends from 2011 to 2019. The term-specific search index is normalized to 100 for its maximum level during this period. The first red line marks April 2013, when Constitutional Amendment 72/13 was enacted, while the second red line indicates June 2015, when Complementary Law 150/15 was implemented.

Figure 2: Labor claims related to domestic work around CA 72 and CL 150



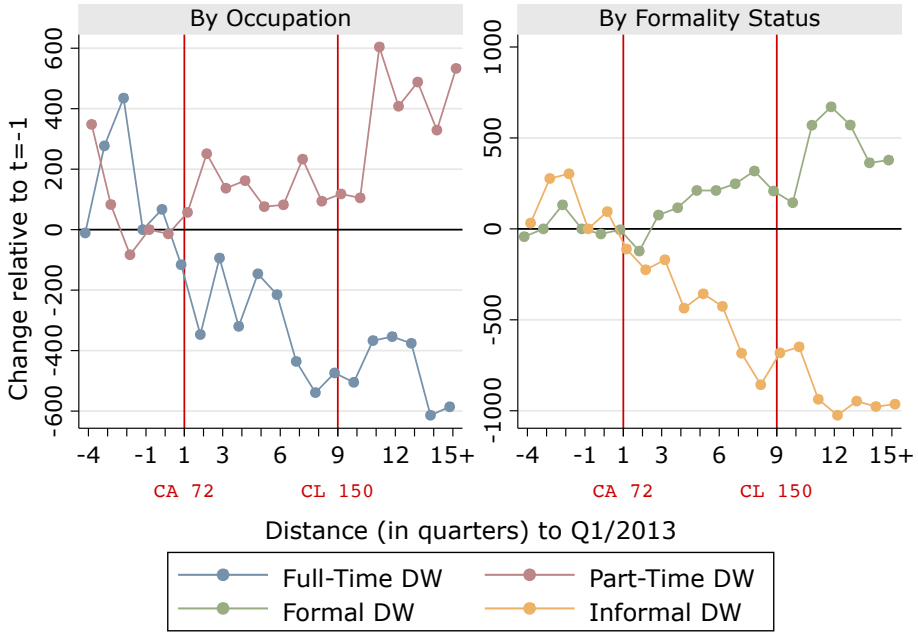
(A) Number of labor claims (national total)



(B) Labor claim index, averaged across regional courts

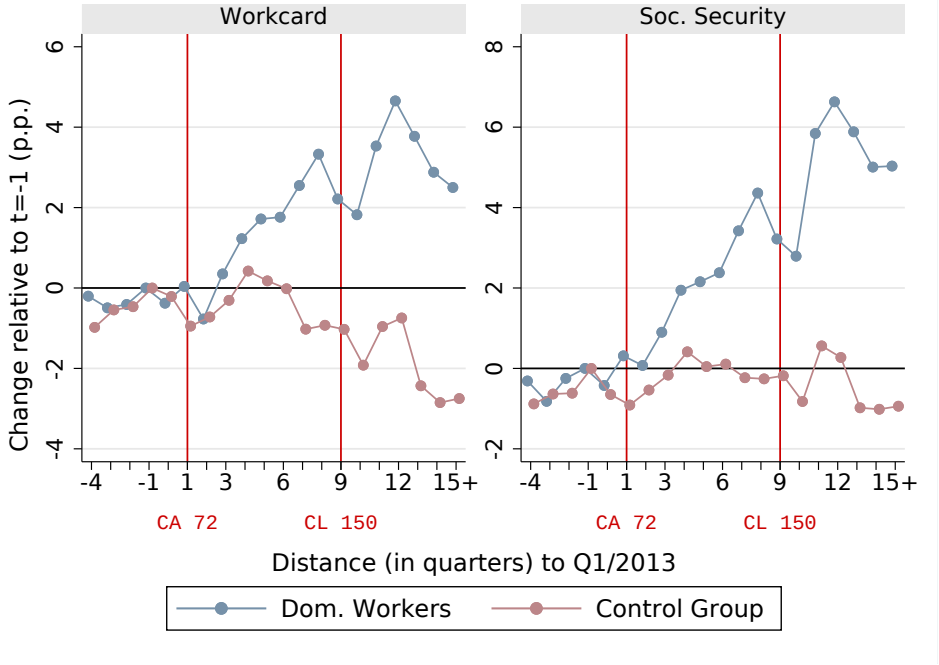
Notes: Yearly data sourced from the Tribunal Superior do Trabalho (TST), "Casos Novos por Atividade Econômica" statistics, covering new cases filed at the trial courts (Varas do Trabalho) under the "Serviços Domésticos" economic activity. The red lines represent April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. Panel A shows the national total of these claims, summing across all 24 regional labor court (TRT) jurisdictions. Panel B first normalizes each region's claim count to its 2012 value (2012 = 100) and then averages these indexes across the 24 TRT jurisdictions, giving each region equal weight.

Figure 3: Time series of number of domestic workers around CA 72 and CL 150



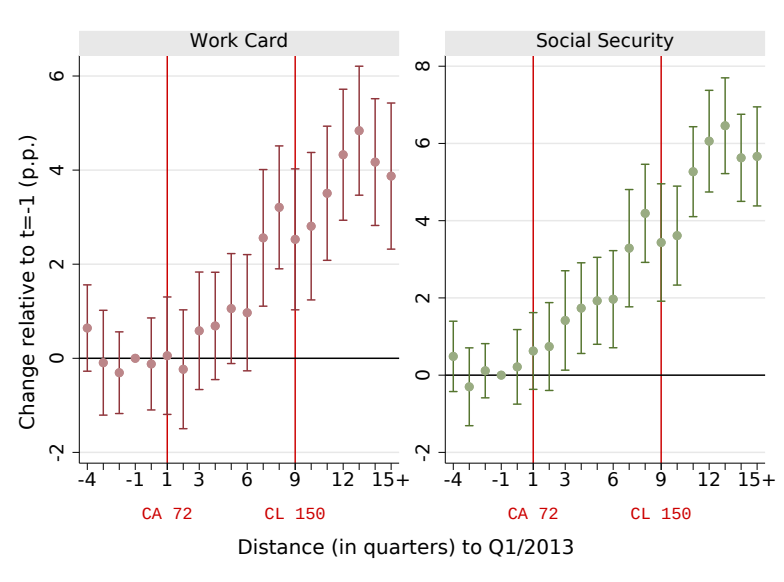
Notes: The graphs split domestic workers by formality status given full-time status (on the right) and full/part-time status (on the left), measuring variations in their numbers in our sample in each quarter, relative to quarter $t = -1$, just before the enactment of CA 72/13. The y-axis is presented in thousands. The red lines correspond to April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 was implemented.

Figure 4: Formality share of treatment and control groups around CA 72 and CL 150

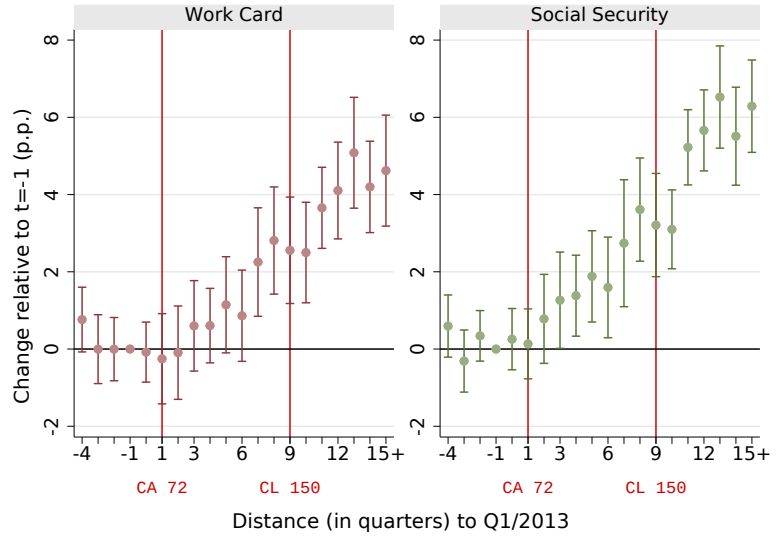


Notes: The graphs display the evolution (in quarters) of formality rates for the treatment group (domestic workers) and the schooling control group, relative to quarter $t = -1$, just before CA 72. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. No control variables are included in this model. Define $trans^{o}_{DW}$. This variable represents the proportion of employees in occupation o who were domestic workers in the previous quarter (quarter $t-1$). In the schooling control group, shown here, we include occupations with a $trans^{o}_{DW}$ value at or below the 25th percentile and an average schooling years at or below the 75th percentile.

Figure 5: Event study effects of CA 72 and CL 150 on formality of domestic workers



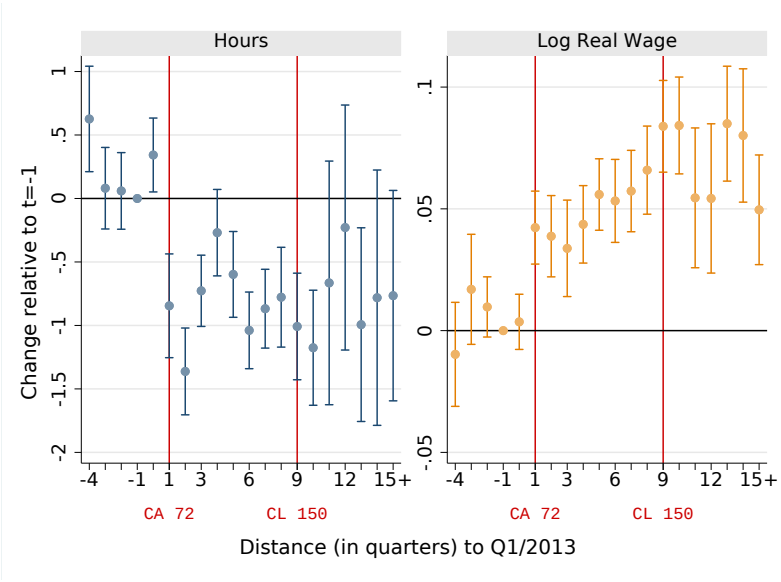
(A) Simple event study



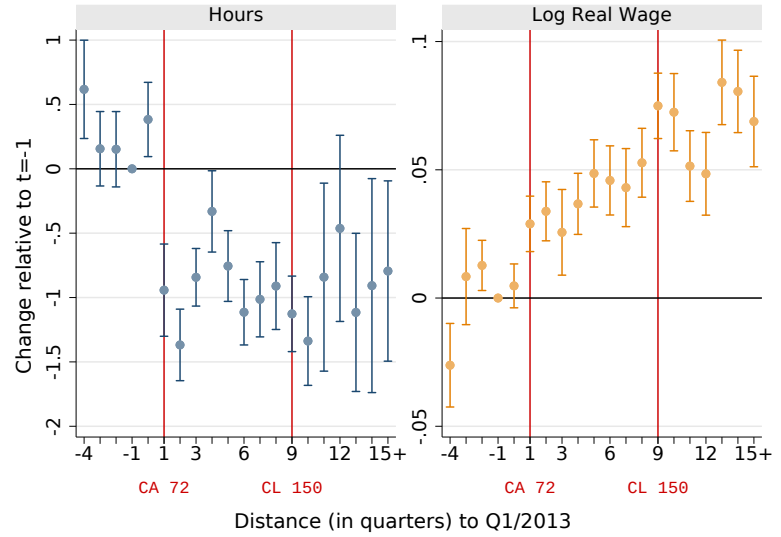
(B) All controls and fixed effects

Notes: The graphs display event study analyses examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' formality outcomes (work card or social security contributions), compared to the control group. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media. The schooling control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. Panel A presents a simple event study, while Panel B includes our covariates for individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and neighborhood.

Figure 6: Event study effects of CA 72 and CL 150 on wages and hours of domestic workers



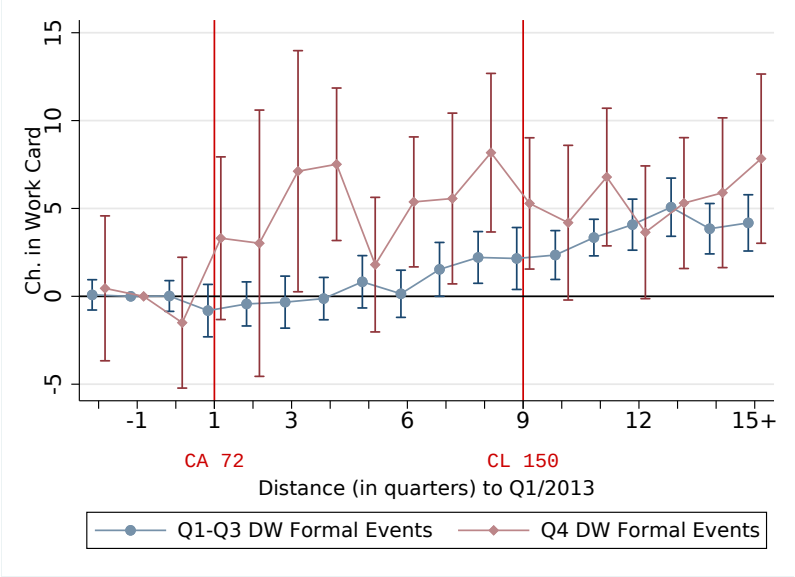
(A) Simple event study



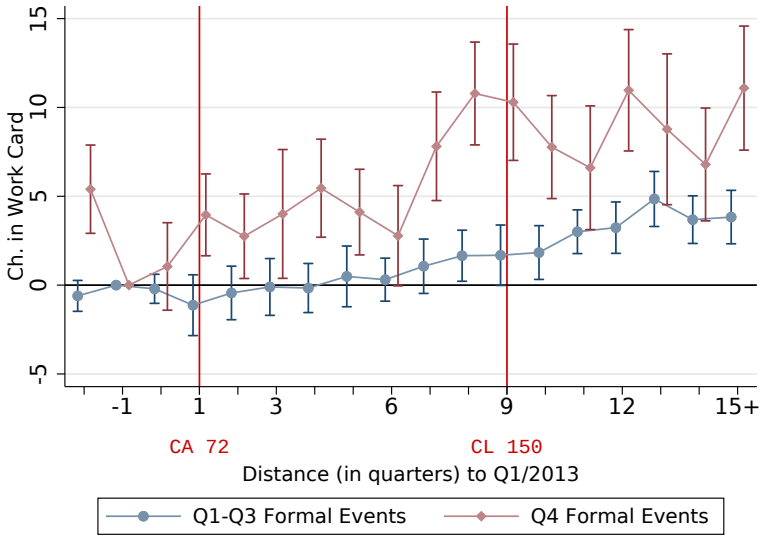
(B) All controls and fixed effects

Notes: The graphs display event study analyses examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' labor market outcomes (either weekly hours worked or log of real monthly wages), compared to the control group. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media. The schooling control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. Panel A presents a simple event study, while Panel B includes our covariates for individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and neighborhood.

Figure 7: Effects of CA 72 and CL 150 on domestic workers' formality by formality events



(A) Number of domestic worker formalizations



(B) Number of non-domestic worker formalizations

Notes: The graphs present event study analyses, examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' formality (work card registration), compared to the control group. The treatment group is separated by quartile of number of neighborhood formalization events, either for domestic workers or non-domestic workers. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media. The schooling control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. Results include our covariates for individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and neighborhood.

Tables

Table 1: Effects of CA 72 and CL 150 on domestic workers' effective labor rights

Labor right	Before	CA 72/13	CL 150/15
Minimum wage	✓	✓	✓
Thirteenth salary	✓	✓	✓
Paid vacation	✓	✓	✓
Maternity leave (4 months)	✓	✓	✓
Soc. Security contribution	✓	✓	✓
Transportation vouchers	✓	✓	✓
Prior notice (minimum of 30 days)	✓	✓	✓
Maximum of 8 daily hours and 44 weekly hours	✗	✓	✓
Overtime bonus of 50%	✗	✓	✓
Nighttime work bonus pay	✗	✗	✓
Severance package	✗	✗	✓
Unemployment benefits	✗	✗	✓
Contribution to FGTS	✗	✗	✓
Family wage	✗	✗	✓
Insurance on work-related accidents	✗	✗	✓

Notes: The table provides an overview of effective labor rights for formal domestic workers in Brazil during three periods: before CA 72/13 (Constitutional Amendment 72/2013), between CA 72/13 and CL 150/15 (Complementary Law 150/2015), and after CL 150/15. Red checkmarks highlight the labor rights introduced or expanded by a given legislation. The FGTS contribution aims to protect workers from significant income shocks. At the beginning of each month, companies deposit an amount equal to 8% of each employee's salary toward this fund. Funds in this account can be used in cases of layoffs, health issues, or to assist in purchasing real estate for housing.

Table 2: Example of effects of CA 72 and CL 150 on costs of formal domestic workers

	Before CA 72/13	After CA 72/13	After CL 150/15
Panel A. Remuneration			
Base salary (44 weekly hours)	R$1,000.00	R$1,000.00	R$1,000.00
Hours above 44 per week	R$90.91	R$136.36	R$136.36
Monthly salary	R$1,090.91	R$1,136.36	R$1,136.36
Thirteenth salary	R$90.91	R$94.70	R$94.70
Vacation bonus	R$30.30	R$31.57	R$31.57
Contribution base	R$1,212.12	R$1,262.63	R$1,262.63
Panel B. Other employer costs			
Transportation voucher (net)	R$94.00	R$94.00	R$94.00
Employer social security (INSS)	R$145.45	R$151.52	R$101.01
Work-accident insurance (SAT)	–	–	R$10.10
FGTS	–	–	R$101.01
Dismissal indemnity fund	–	–	R$40.40
Total monthly cost	R$1,451.57	R$1,508.15	R$1,609.15
Change from before CA 72/13 (%)	–	3.9	10.9

Notes: The table reports the monthly cost of employing one formal domestic worker under the rules in force before CA 72/13 (Constitutional Amendment 72/2013), between CA 72/13 and CL 150/15 (Complementary Law 150/2015), and after CL 150/15. The example assumes a monthly base salary of R$1,000 for a 44-hour workweek, 4 additional hours per week, and monthly transportation costs of R$154. The hourly wage equals the base salary divided by 220, the number of monthly hours that Brazilian labor law assigns to a 44-hour workweek. The 4 overtime weekly hours correspond to 20 monthly hours. Before CA 72/13, domestic workers had no limit on weekly hours, so additional hours are paid at the regular hourly wage. After CA 72/13, they are paid at a 50% premium. The thirteenth salary (one extra monthly salary paid each year) and the vacation bonus (one-third of a monthly salary paid each year) are spread evenly across the 12 months. Payroll contributions are levied on monthly remuneration plus these two items, reported as the contribution base. The worker pays transportation costs up to 6% of the base salary, and the employer pays the remainder. The employer social security contribution equals 12% of the contribution base before CL 150/15 and 8% after. CL 150/15 also requires employer contributions of 8% to the FGTS (Length-of-Service Guarantee Fund), a mandatory savings account in the worker's name; 0.8% to work-accident insurance (SAT); and 3.2% to a dismissal indemnity fund. The table reports statutory employer costs at a fixed gross wage. It excludes the employee social security contribution, which the law assigns to the worker, and income tax, which is zero at this wage.

Table 3: Summary statistics

	Domestic Workers	All Other Workers
Formality (Work Card) (%)	29.1	69.5
	(45.4)	(46.0)
Formality (Soc. Security) (%)	34.9	75.5
	(47.7)	(43.0)
Weekly Hours	33.68	40.12
	(16.63)	(12.97)
Monthly Wage	R$518.1	R$1,172.5
	(R$379.4)	(R$1,487.7)
Years of Schooling	6.99	9.90
	(3.68)	(4.11)
White (%)	33.4	44.8
	(47.2)	(49.7)
Woman (%)	91.5	36.4
	(27.9)	(48.1)
Age	39.32	34.03
	(12.41)	(11.89)
Observations	64,799	495,102

Notes: The table provides descriptive statistics in our dataset for domestic workers and non-domestic worker employees in Brazil prior to the implementation of CA 72/13 (Constitutional Amendment 72/2013) in 2012. The statistics include two measures of formality (work card registration with the government and employee social security contributions). Standard deviations are shown in parentheses below the mean values for each variable. The total number of observations for domestic workers and other workers is given at the bottom of the table. The data comes from the PNAD Continua rotating panel dataset.

Table 4: Top occupations in schooling control group

Occupation	Count	Share(%)
Secretarial supervisors	19,727	5.5
Agricultural and industrial machinery mechanics	17,513	4.9
Machine tool regulators and operators	15,490	4.3
Welders and oxy-fuel cutting	14,491	4.0
Couriers, baggage handlers and parcel couriers	14,173	3.9
Military graduates and squares	13,921	3.9
Wholesale and retail managers	13,790	3.8
Trade representatives	13,623	3.8
Gas station attendants	13,561	3.8
Auto bodyworkers	11,500	3.2
Electrotechnicians	10,343	2.9
Joiners and the like	9,982	2.8
Total	359,281	100.0

The table shows the distribution of occupations in the schooling control group, ranked by the number of times they were included in this control group. The data comes from the PNAD Continua rotating panel dataset spanning 2012 to 2016. Define $trans^{o}_{DW}$. This variable represents the proportion of employees in occupation o who were domestic workers in the previous quarter (quarter $t-1$). In the schooling-based control group, we include occupations with a $trans^{o}_{DW}$ value at or below the 25th percentile and an average years of schooling at or below the 75th percentile among all occupations.

Table 5: Triple differences estimates of peers on CA 72 and CL 150 effects

	Other Dom Workers	Non-HH Dom Workers	Mean Other DW
Post CA/Pre CL	0.91***	0.82***	1.20***
	(0.17)	(0.17)	(0.22)
Post CL	0.21	0.11	0.54**
	(0.21)	(0.20)	(0.25)
Observations	473374	473374	449753

	Other Formal Workers	Other Formal DWs
Post CA/Pre CL	-0.047	-0.42
	(0.045)	(0.92)
Post CL	-0.14	-2.71
	(0.098)	(1.88)
Observations	473374	471146

(A) Work Card Registration

	Other Dom Workers	Non-HH Dom Workers	Mean Other DW
Post CA/Pre CL	0.74***	0.65***	1.02***
	(0.16)	(0.16)	(0.22)
Post CL	0.27	0.16	0.50**
	(0.21)	(0.21)	(0.25)
Observations	480635	480635	456641

	Other Formal Workers	Other Formal DWs
Post CA/Pre CL	0.12**	2.03**
	(0.048)	(0.89)
Post CL	0.19**	1.55
	(0.081)	(1.80)
Observations	480635	478281

(B) Social Security Contributions

Notes: Standard errors clustered by occupation in parentheses. Stars denote statistical significance (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). The table shows the triple differences estimation relating the difference-in-differences effects of CA 72 and CL 150 on domestic workers with peer effect variables. Two periods are considered: "Post CA/Pre CL" (after CA 72 but before CL 150) and "Post CL" (after CL 150). The presented estimates are interacted with the treatment group dummy and the peer variable; other triple differences coefficients are omitted. Covariates include individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood – or UPA – level) of years of schooling, age, real wages, female, and white. Fixed effects are included for occupation, date, and neighborhood. The dependent variables are: Work Card (work permit definition of formality) in Panel A and Soc. Security (social security contribution definition of formality) in Panel B. The variables interacted with the DID estimator include: Other Dom Workers (number of other domestic workers in the neighborhood), Non-HH Dom Workers (number of non-household domestic workers in the neighborhood), Mean Other DW (mean number of other domestic workers in the neighborhood), Other Formal Workers (number of other formal workers in the neighborhood), and Other Formal DWs (number of other formal domestic workers in the neighborhood).

Table 6: Triple differences estimates of formalization events on CA 72 and CL 150 effects

	Formal Events	DW Formal Events
Post CA/Pre CL	0.45	8.54***
	(0.70)	(1.90)
Post CL	0.40	3.58
	(0.96)	(2.18)
Observations	391895	391895

(A) Work Card Registration

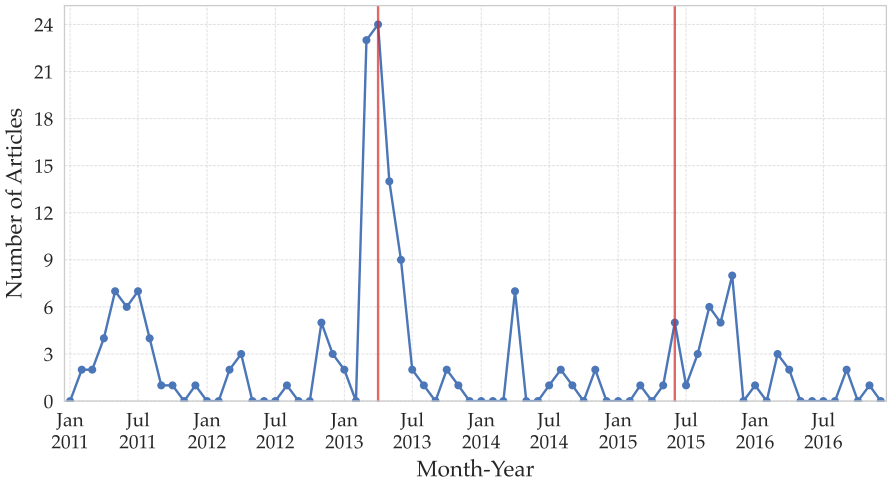
	Formal Events	DW Formal Events
Post CA/Pre CL	1.00	8.82***
	(0.61)	(2.04)
Post CL	0.49	4.26**
	(0.83)	(1.84)
Observations	397970	397970

(B) Social Security Contributions

Notes: Standard errors clustered by occupation in parentheses. Stars denote statistical significance (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). The table shows the triple differences estimation relating the difference-in-differences effects of CA 72 and CL 150 on domestic workers with formalization events variables. Two periods are considered: "Post CA/Pre CL" (after CA 72 but before CL 150) and "Post CL" (after CL 150). The presented estimates are interacted with the treatment group dummy and the formalization event variable; other triple differences coefficients are omitted. Covariates include individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood – or UPA – level) of years of schooling, age, real wages, female, and white. Fixed effects are included for occupation, date, and neighborhood. The dependent variables are: Work Card (work permit definition of formality) in Panel A and Soc. Security (social security contribution definition of formality) in Panel B. Variables interacted with the DID estimator include: number of other domestic workers (DW Formal Events) and non-domestic workers (Formal Events) who became formal in the neighborhood in the past 2 quarters.

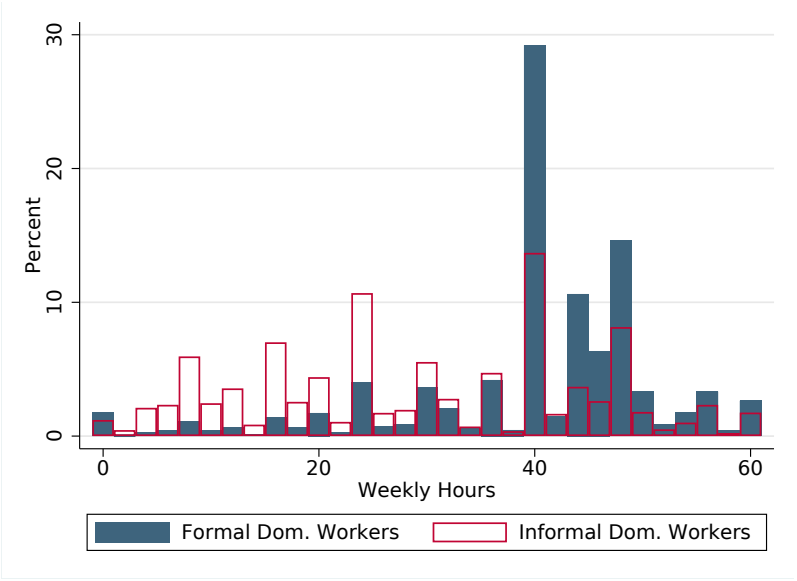
Appendix A: Additional Figures and Tables

Figure A1: Newspaper articles related to domestic workers around CA 72 and CL 150

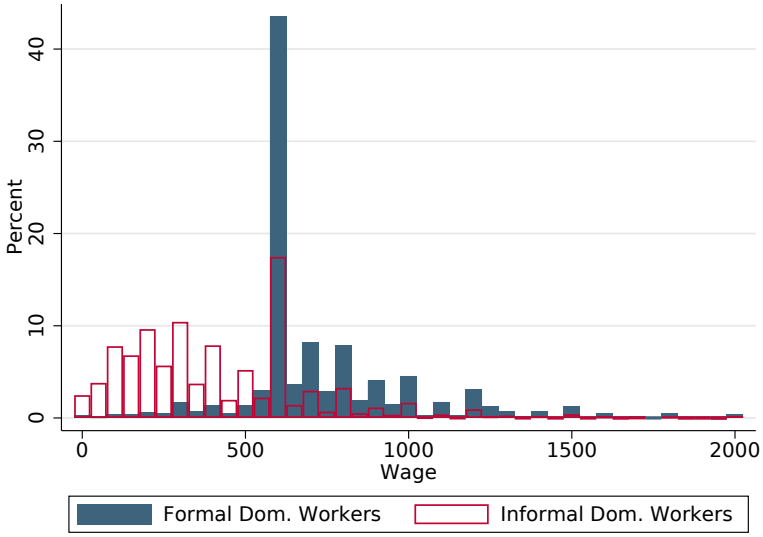


Notes: Monthly data from Estado de São Paulo newspaper from 2011 to 2016. All articles mentioning domestic workers in the headline are included. The first red line marks April 2013, when Constitutional Amendment 72/13 was enacted, while the second red line indicates June 2015, when Complementary Law 150/15 was implemented.

Figure A2: Histogram for domestic workers' hours and wages before CA 72/13



(A) Weekly hours worked



(B) Monthly wages

Notes: The graphs show the average of quarterly data from 2012 using the rotating panel dataset PNAD Contínua. The first histogram illustrates the distribution of weekly hours worked by domestic workers in 2012, before the implementation of CA 72/13 (Constitutional Amendment 72/2013). The second histogram displays the distribution of real monthly wages during the same period. Both histograms are divided into formal and informal domestic worker categories. Formal workers have a work card registration or social security contributions, while informal workers do not. The maximum wage is capped at R$2000,00 in the second graph.

Figure A3: Event study effects of CA 72 and CL 150 on other outcomes of domestic workers



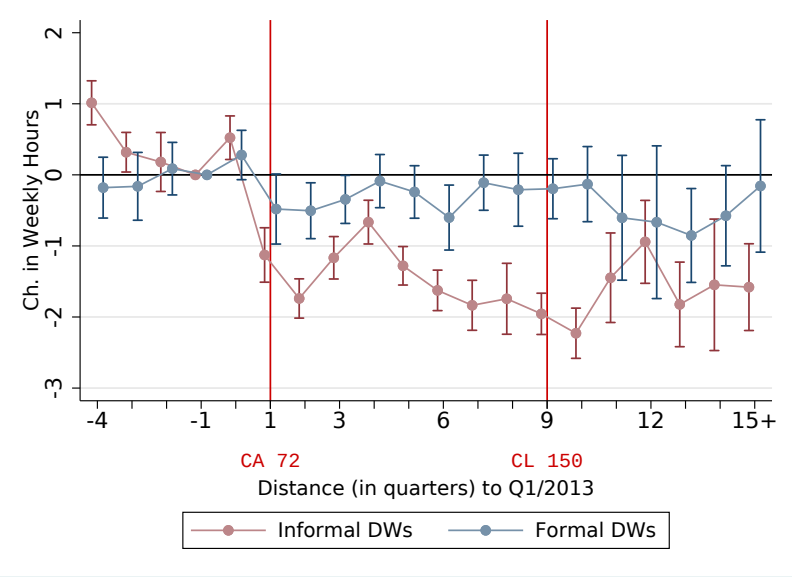
(A) Simple event study



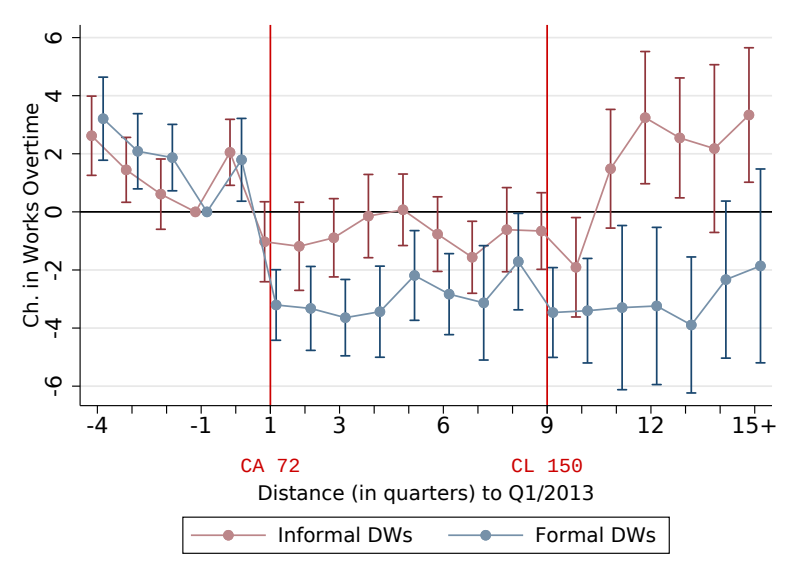
(B) All controls and fixed effects

Notes: The graphs display event study analyses examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' labor market outcomes (binary variables for working overtime or getting paid the minimum wage), compared to the control group. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media. The schooling control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. Panel A presents a simple event study, while Panel B includes our covariates for individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and neighborhood.

Figure A4: Event study results by formality status of domestic workers

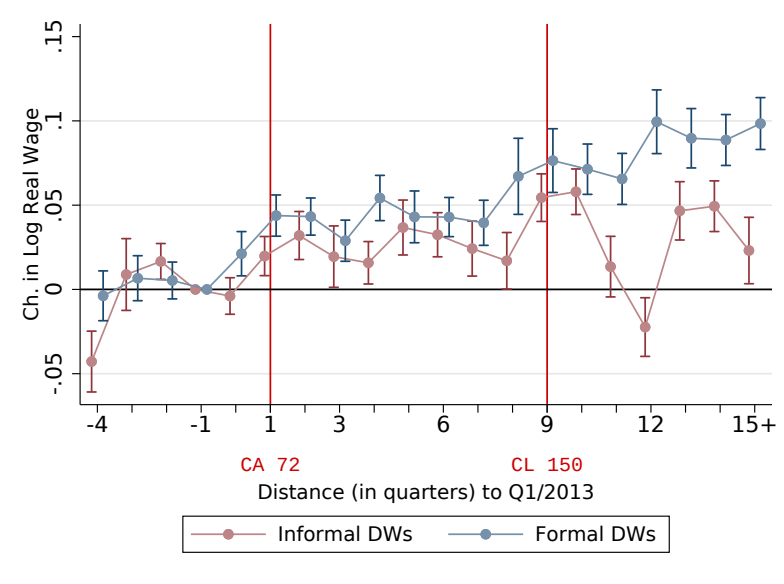


(A) Weekly hours

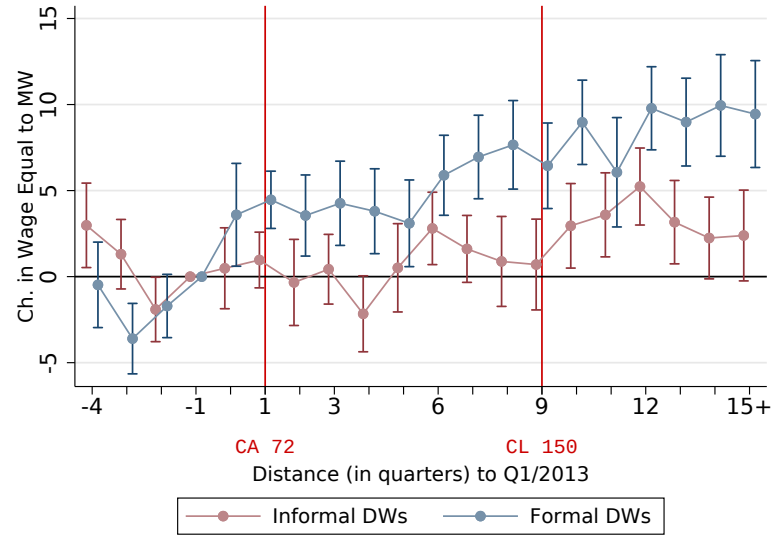


(B) Works overtime

Figure A4: Event study results by formality status of domestic workers (cont.)



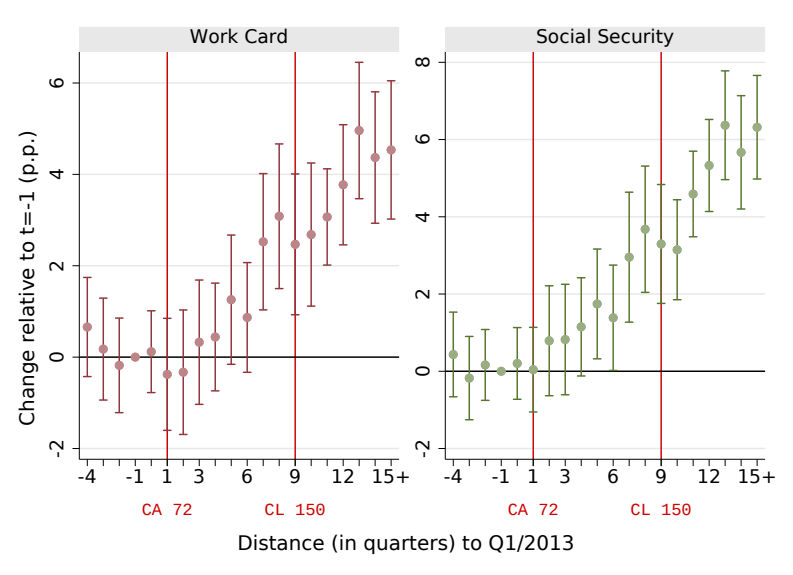
(C) Log of real monthly wages



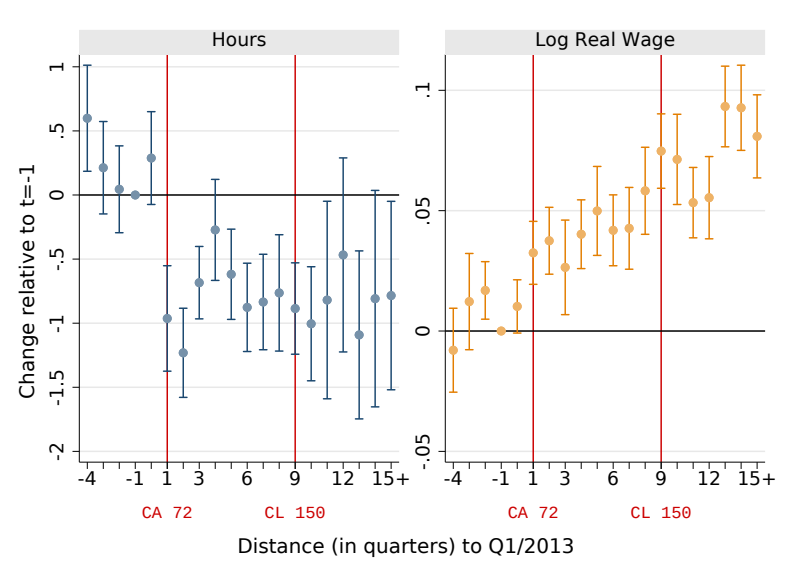
(D) Paid the minimum wage

Notes: The graphs display event study analyses examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' labor market outcomes, separately by formality status, compared to the control group. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 came into effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media. The schooling control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. We include our covariates for individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and neighborhood.

Figure A5: Effects of CA 72 and CL 150 on domestic workers (wage control group)



(A) Formality



(B) Weekly hours and real wages

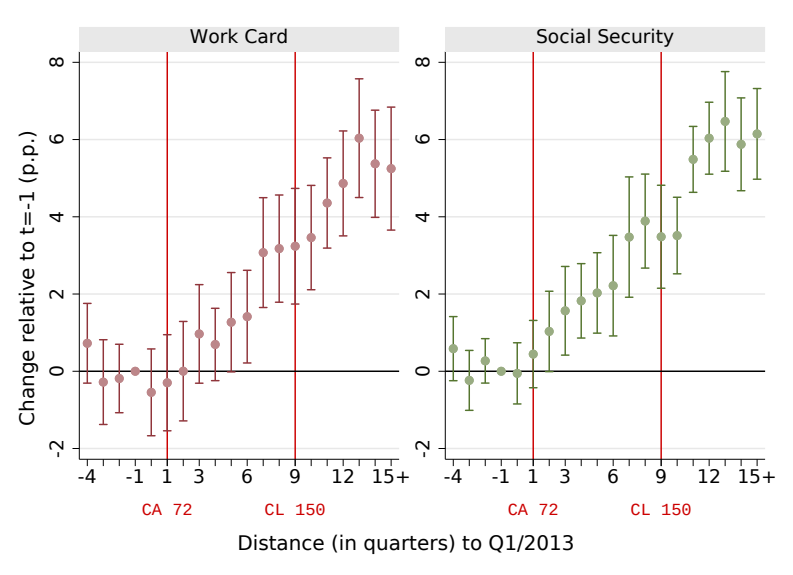
Figure A5: Effects of CA 72 and CL 150 on domestic workers (wage control group) (cont.)



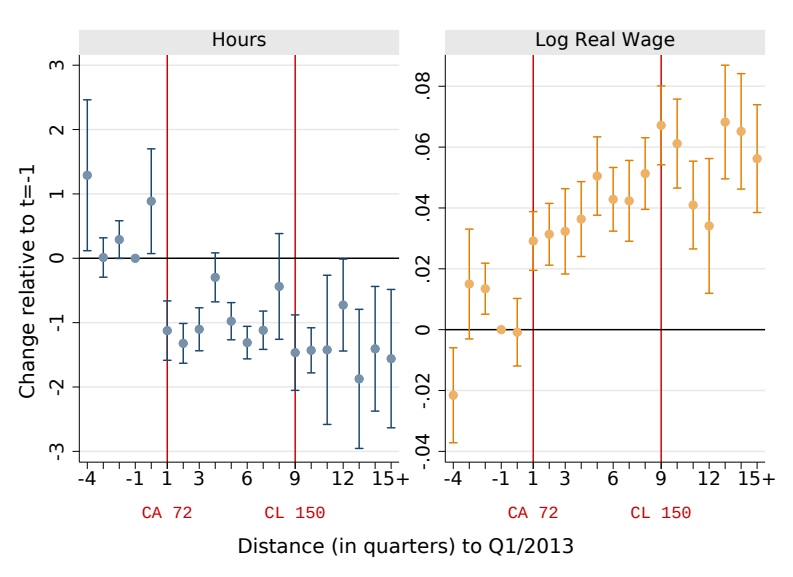
(C) Works overtime and is paid the minimum wage

Notes: The graphs display event study analyses examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' labor market outcomes, compared to the control group. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media. The wage control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. Results include our covariates for individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and neighborhood.

Figure A6: Effects of CA 72 and CL 150 on domestic workers (race control group)

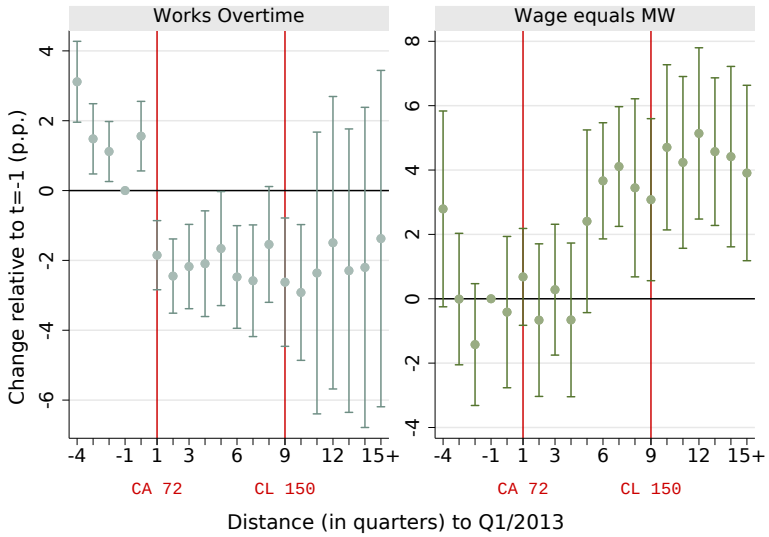


(A) Formality



(B) Weekly hours and real wages

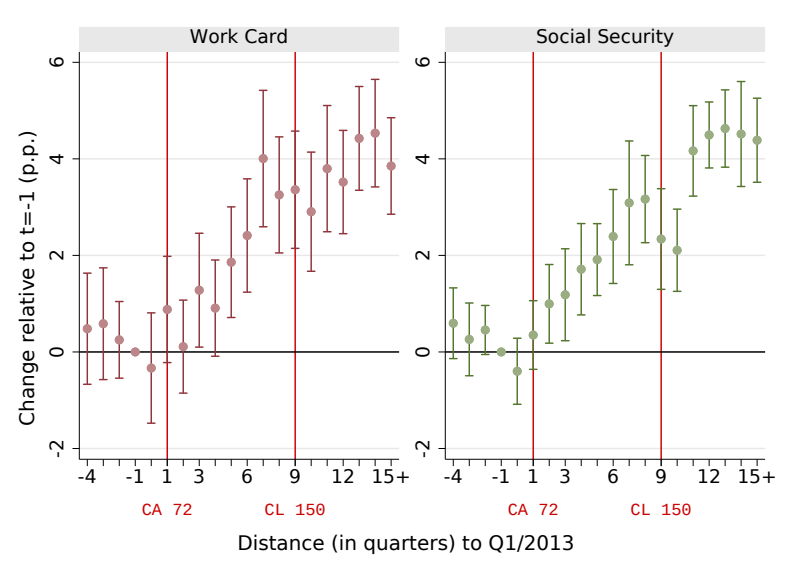
Figure A6: Effects of CA 72 and CL 150 on domestic workers (race control group) (cont.)



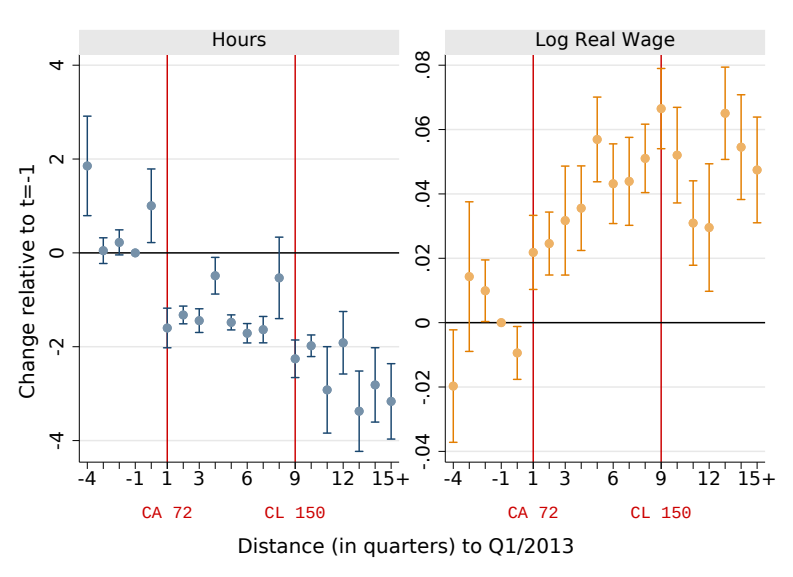
(C) Works overtime and is paid the minimum wage

Notes: The graphs display event study analyses examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' labor market outcomes, compared to the control group. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media. The race control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. Results include our covariates for individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and neighborhood.

Figure A7: Effects of CA 72 and CL 150 on domestic workers (gender control group)



(A) Formality



(B) Weekly hours and real wages

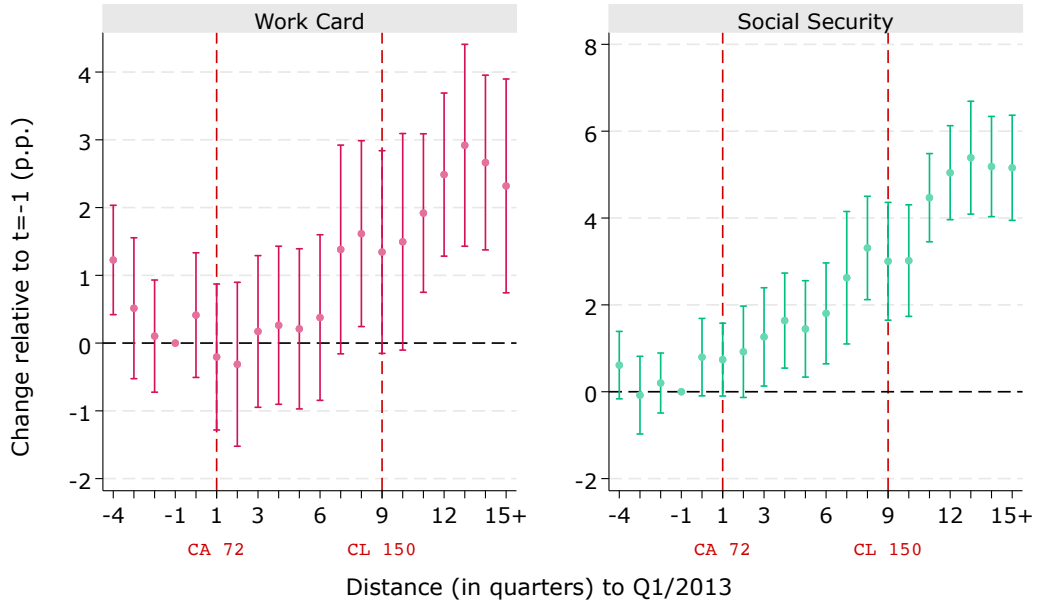
Figure A7: Effects of CA 72 and CL 150 on domestic workers (gender control group) (cont.)



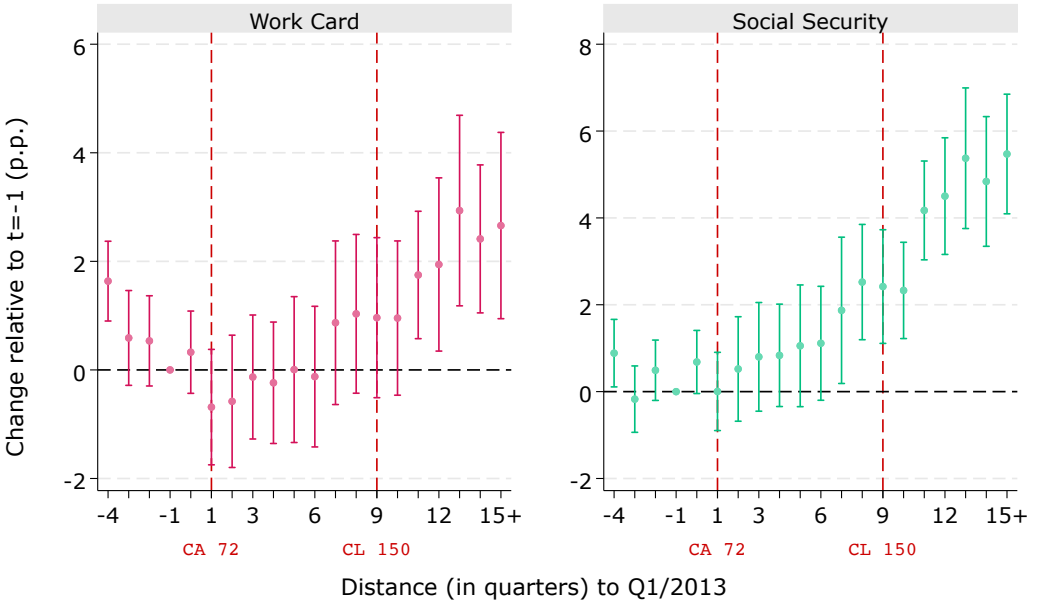
(C) Works overtime and is paid the minimum wage

Notes: The graphs display event study analyses examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' labor market outcomes, compared to the control group. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media. The gender control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. Results include our covariates for individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and neighborhood.

Figure A8: Event study effects on formality of domestic workers (with diaristas)



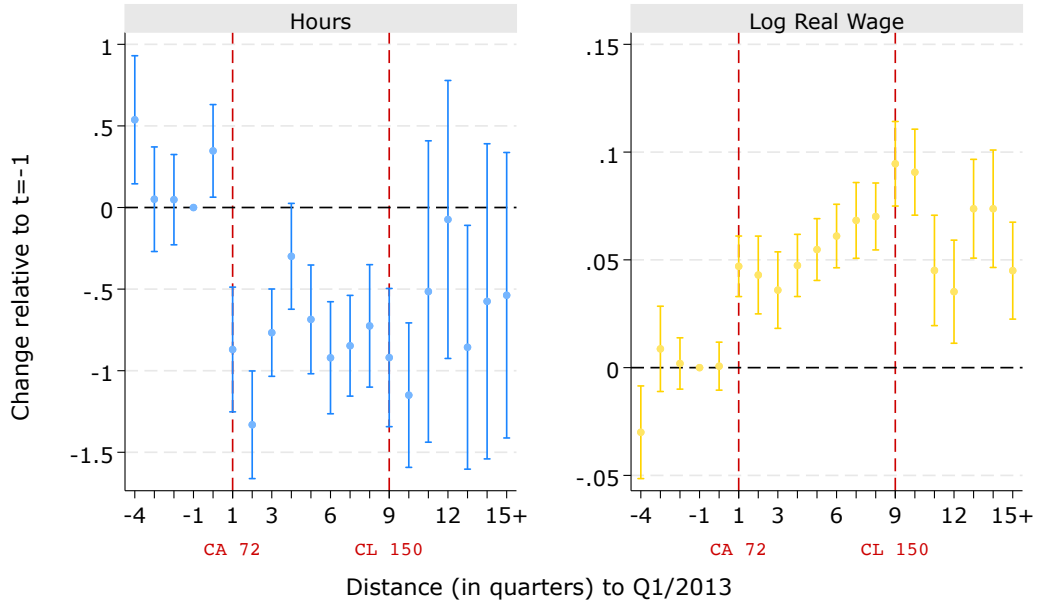
(A) Simple event study



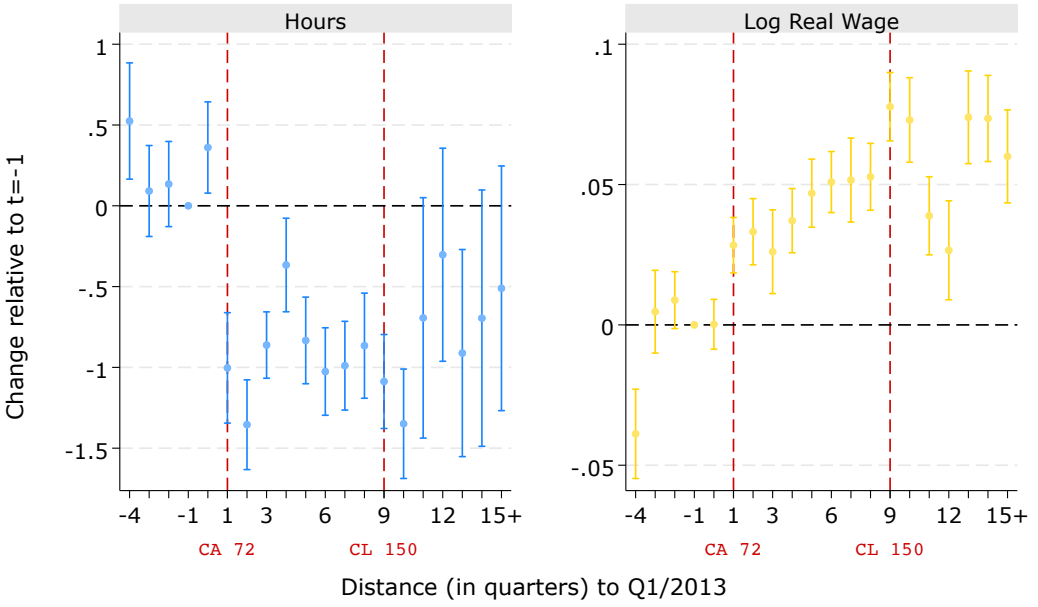
(B) All controls and fixed effects

Notes: The graphs display event study analyses examining the effects of CA 72 and CL 150 on domestic workers' formality outcomes (work card or social security contributions), compared to the control group. Unlike the baseline specification, the sample includes part-time domestic workers (*diaristas*). The red lines mark April 2013 and June 2015, when CA 72 and CL 150 took effect, respectively. The base period is $t = -1$ (Q4/2012). The schooling control group is used. Lines around point estimates show 95% confidence intervals, with standard errors clustered by occupation. Panel A presents a simple event study, while Panel B includes individual and neighborhood-level controls and occupation, quarter, and neighborhood fixed effects, as described in the main specification.

Figure A9: Event study effects on wages and hours of domestic workers (with diaristas)



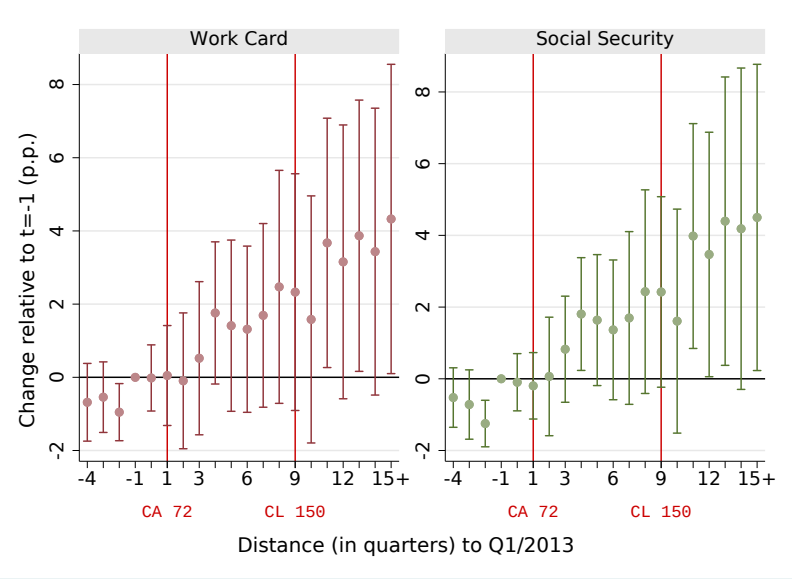
(A) Simple event study



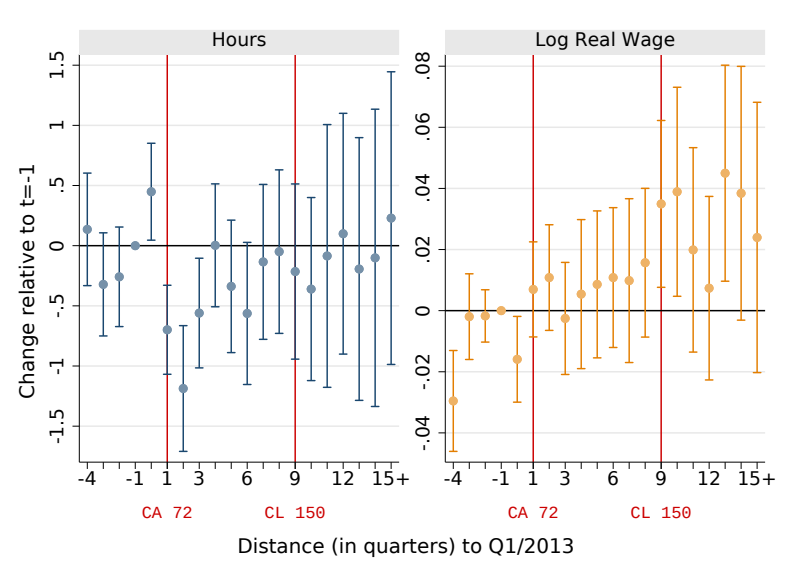
(B) All controls and fixed effects

Notes: The graphs display event study analyses examining the effects of CA 72 and CL 150 on domestic workers' labor market outcomes (weekly hours worked and log real monthly wages), compared to the control group. Unlike the baseline specification, the sample includes part-time domestic workers (*diaristas*). The red lines mark April 2013 and June 2015, when CA 72 and CL 150 took effect, respectively. The base period is $t = -1$ (Q4/2012). The schooling control group is used. Lines around point estimates show 95% confidence intervals, with standard errors clustered by occupation. Panel A presents a simple event study, while Panel B includes individual and neighborhood-level controls and occupation, quarter, and neighborhood fixed effects, as described in the main specification.

Figure A10: Effects of CA 72 and CL 150 on domestic workers (individual fixed effects)

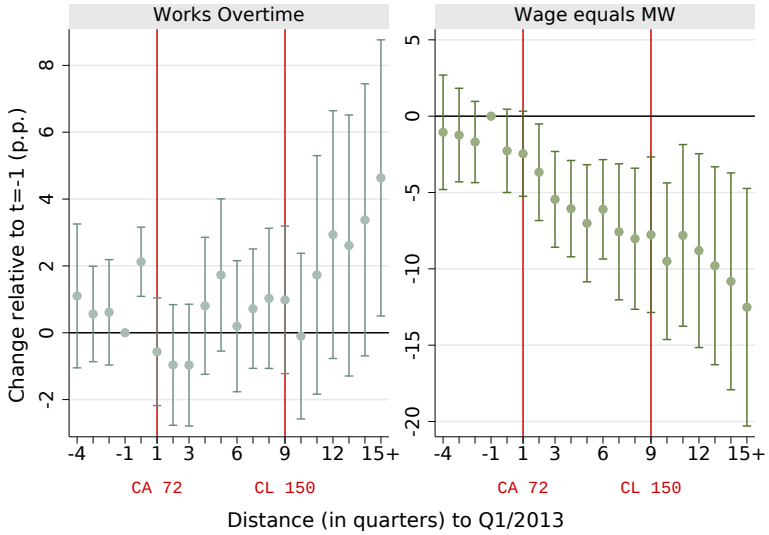


(A) Formality



(B) Weekly hours and real wages

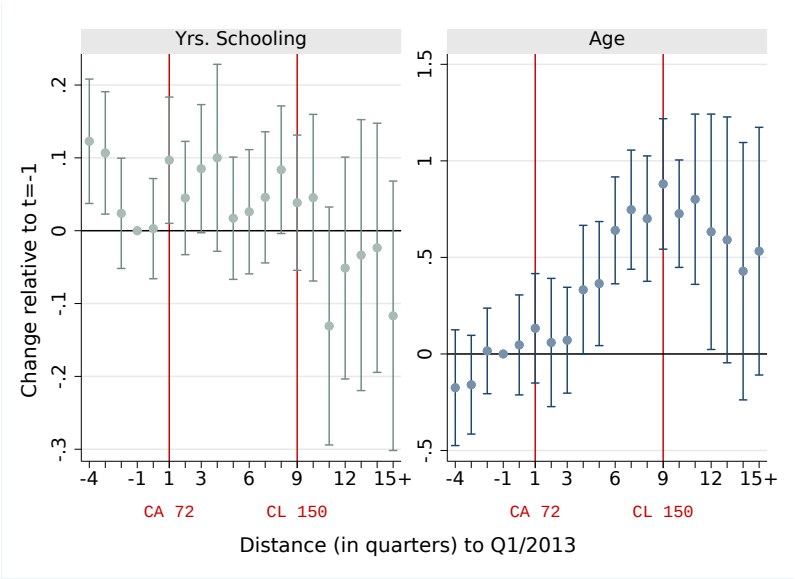
Figure A10: Effects of CA 72 and CL 150 on domestic workers (individual fixed effects) (cont.)



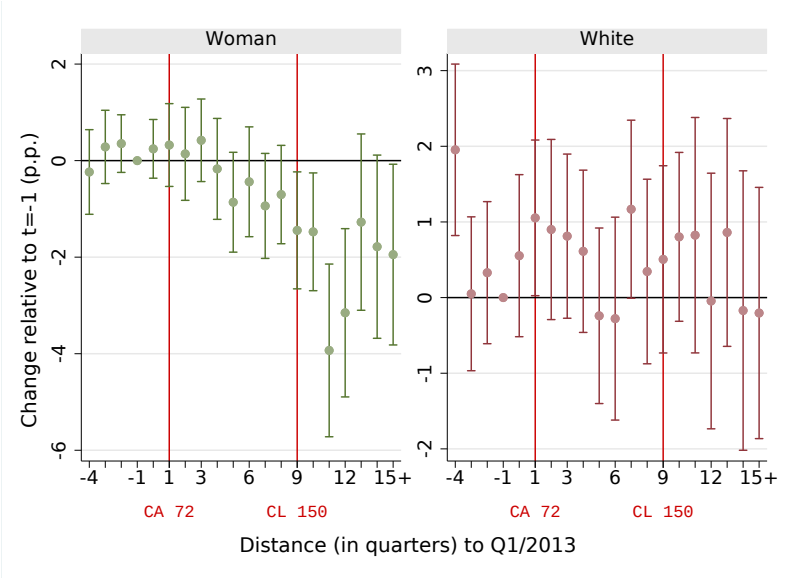
(C) Works overtime and is paid the minimum wage

Notes: The graphs display event study analyses examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' labor market outcomes, compared to the control group. This analysis includes individual fixed effects. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media. The schooling control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. Results include our covariates for individual-level variables: third degree polynomials of schooling and age. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and individual.

Figure A11: Effects of CA 72 and CL 150 on domestic workers' demographic characteristics



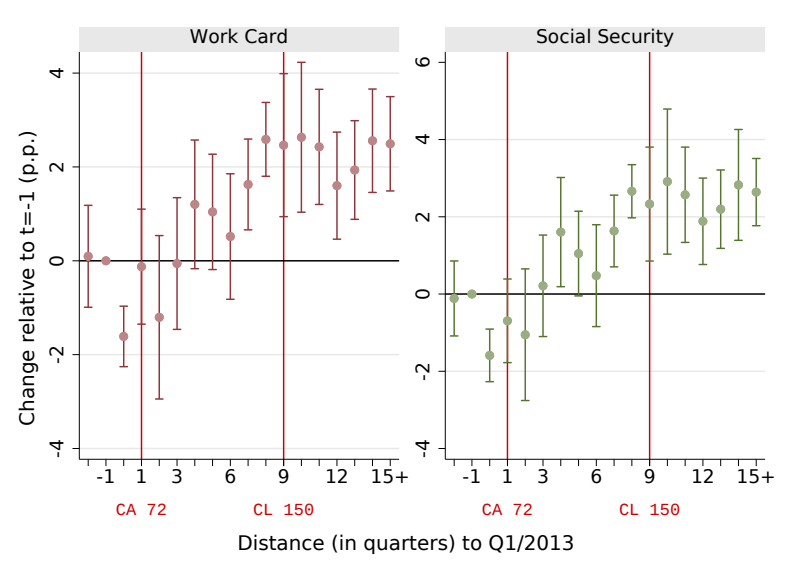
(A) Years of schooling and age



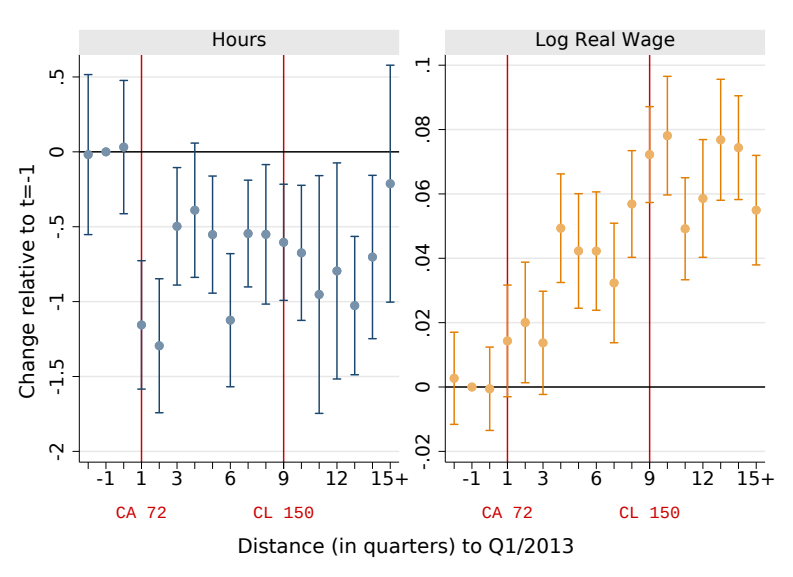
(B) Female and white

Notes: The graphs display event study analyses examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' demographic characteristics, compared to the control group. Panel A reports effects on years of schooling and age. Panel B reports effects on binary variables for whether the worker is female and whether the worker is white. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media. The schooling control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. The estimates are from the simple event study specification without additional covariates or fixed effects.

Figure A12: Effects of CA 72 and CL 150 on domestic workers who were previously formal

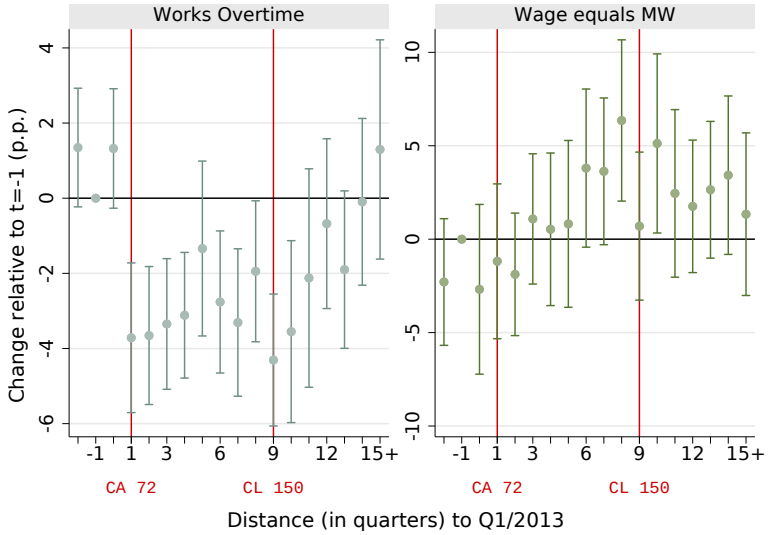


(A) Formality



(B) Weekly hours and real wages

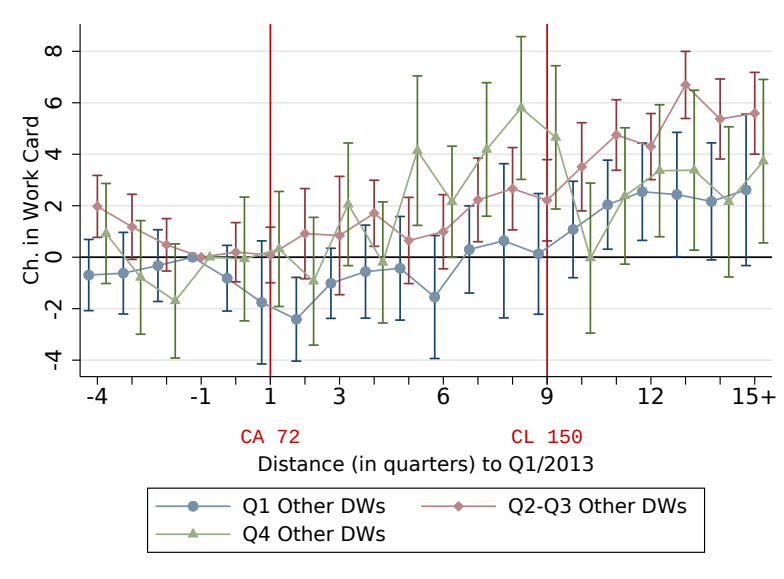
Figure A12: Effects of CA 72 and CL 150 on domestic workers who were previously formal (cont.)



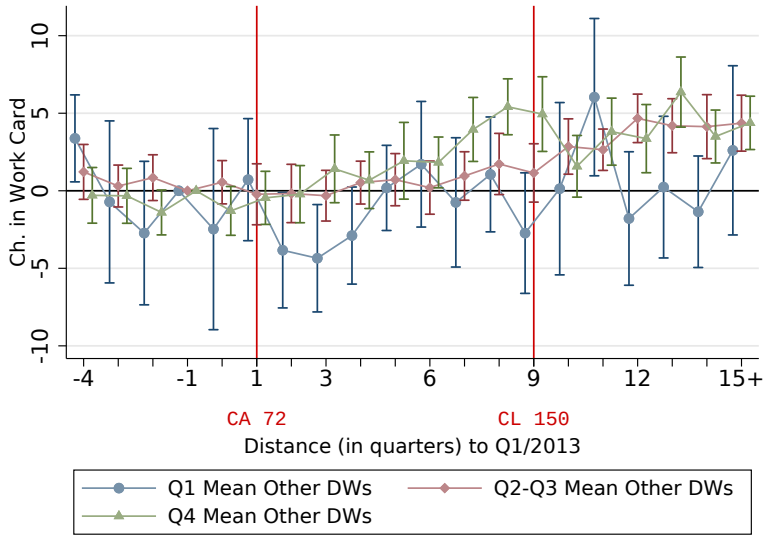
(C) Works overtime and is paid the minimum wage

Notes: The graphs display event study analyses examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' labor market outcomes, compared to the control group. Only workers who were previously formal are included in the treatment group. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media. The schooling control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. Results include our covariates for individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and neighborhood.

Figure A13: Effects of CA 72 and CL 150 on domestic workers' formality by peers quartile



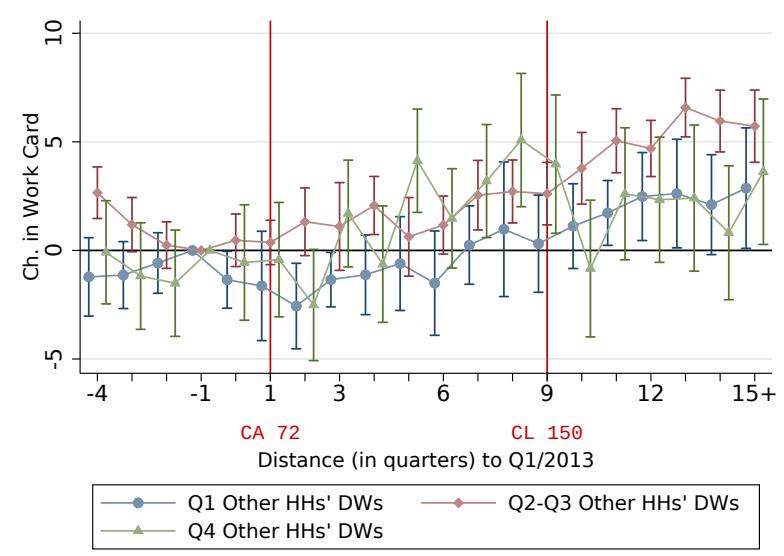
(A) Number of other domestic workers



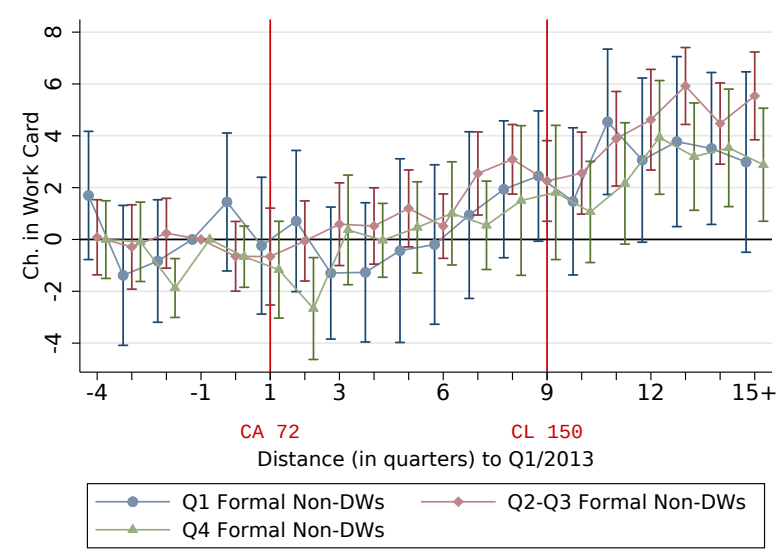
(B) Average number of other domestic workers (3 quarters)

Notes: The graphs present event study analyses, examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' formality (work card registration), compared to the control group. The treatment group is separated by quartile of number of neighborhood peers. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media.The schooling control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. Results include our covariates for individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and neighborhood.

Figure A14: Effects of CA 72 and CL 150 on domestic workers' formality by quartile of other groups of peers

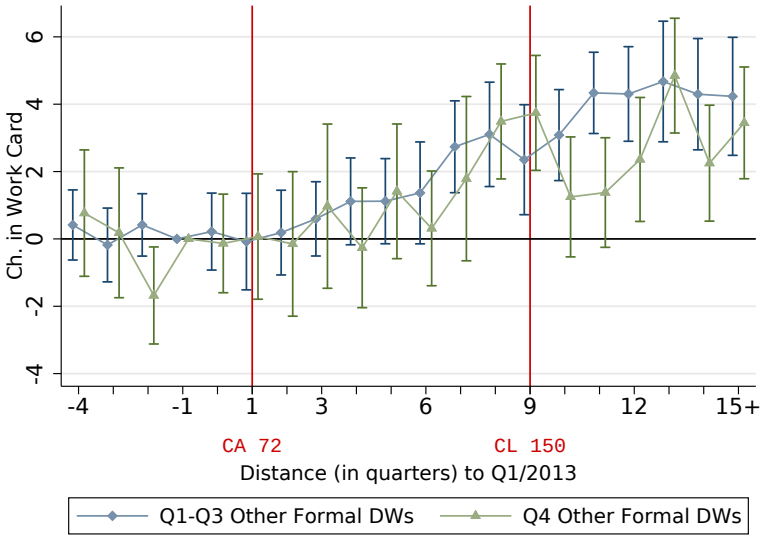


(A) Number of other DWs outside household



(B) Number of formal non-DWs

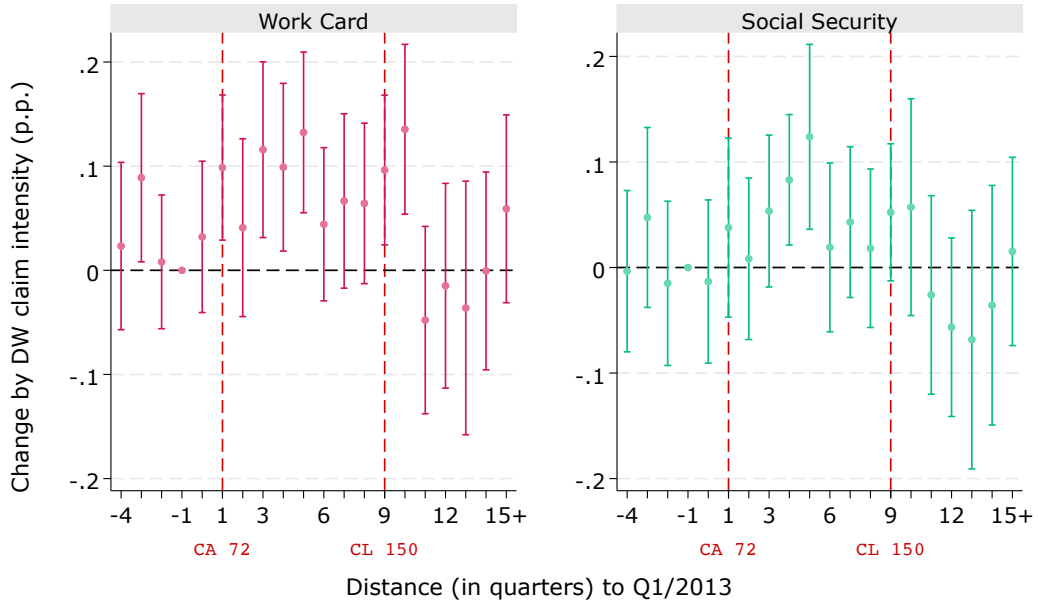
Figure A14: Effects of CA 72 and CL 150 on domestic workers' formality by quartile of other groups of peers (cont.)



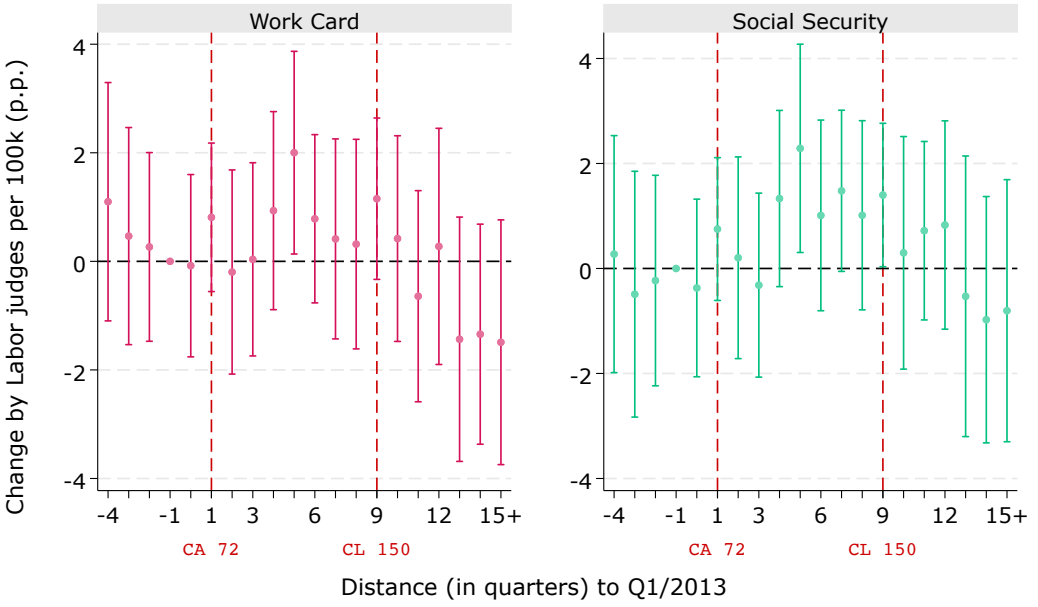
(C) Number of formal DWs

Notes: The graphs present event study analyses, examining the effects of CA 72 (Constitutional Amendment 72) and CL 150 (Complementary Law 150) on domestic workers' formality (work card registration), compared to the control group. The treatment group is separated by quartile of number of neighborhood peers. The red lines mark April 2013, when Constitutional Amendment 72/13 was enacted, and June 2015, when Complementary Law 150/15 took effect. The base period for the event study is $t = -1$, the last period before the CA was discussed heavily in the media.The schooling control group is used in this analysis. The 95% confidence intervals are shown by the lines around point estimates, with standard errors clustered by occupation. Results include our covariates for individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood, or UPA, level) of years of schooling, age, real wages, female, and white, and fixed effects for occupation, quarter, and neighborhood.

Figure A15: Effects of the legislation on domestic workers' formality by enforcement access



(A) Domestic-work labor claims per worker



(B) Labor-court magistrates per 100k inhabitants

Notes: The graphs present triple-differences event study estimates of the effects of CA 72 and CL 150 on domestic workers' formality measures. Panel A interacts the treatment effect with 2012 domestic work labor claims (TST "Casos Novos por Atividade Econômica," Serviços Domésticos) per member of the state's labor force. Panel B interacts it with 2012 first-instance labor court magistrates per 100,000 inhabitants (Relatório Geral da Justiça do Trabalho). Plotted coefficients are the interaction terms. The red lines mark when CA 72/13 and CL 150/15 took effect. The base period is $t = -1$. The schooling control group is used. The 95% confidence intervals are shown, with standard errors clustered by occupation. Both panels include individual-level covariates, third-degree polynomials of the UPA-level leave-out means and fixed effects for occupation, quarter, and neighborhood.

Table A1: Top occupations in treated group

Occupation	Count	Share(%)
Domestic service workers in general	267,437	84.0
Child caretakers	24,019	7.5
Home care workers	16,109	5.1
Cooks	4,781	1.5
Elementary gardening and horticulture workers	2,267	0.7
Drivers of cars, taxis and trucks	2,137	0.7
Farmers and skilled workers in gardens	841	0.3
Housekeepers and domestic butlers	518	0.2
Security guards	401	0.1
Captains, cover officers and practitioners	31	0.0
Water carriers and firewood collectors	8	0.0
Total	318,549	100.0

Notes: The table displays the most common occupations in the treatment group, ranked by the number of workers in each occupation in our dataset. The data comes from the PNAD Continua rotating panel dataset, spanning 2012 to 2016.

Table A2: Distribution of inclusion to control groups by non-domestic worker

Count of control groups inclusion	Count	Share(%)
1	434,992	53.1
2	179,198	21.9
3	194,773	23.8
4	9,763	1.2
Total	818,726	100.0

Notes: The table displays the distribution of non-domestic worker employees in our sample based on the number of control groups they qualify for. The data comes from the PNAD Continua rotating panel dataset spanning 2012 to 2016. The control groups are constructed as follows. Define $trans^{o}_{DW}$. This variable represents the proportion of employees in occupation o who were domestic workers in the previous quarter (quarter $t-1$). For the race control group, we include occupations at or below the 25th percentile of the $trans^{o}_{DW}$ variable and at or below the 75th percentile of the share of white workers among all occupations. The wage control group consists of occupations with a $trans^{o}_{DW}$ value at or below the 25th percentile and an average wage at or below the 75th percentile. In the schooling control group, we include occupations with a $trans^{o}_{DW}$ value at or below the 25th percentile and an average schooling at or below the 75th percentile. The women-focused control group includes occupations that meet the following criteria: (i) at or below the 50th percentile of the $trans^{o}_{DW}$ value; (ii) more than 50% female; and (iii) the individual is female.

Table A3: Top occupations in alternative control groups

Occupation	Count	Share(%)
Elementary school teachers	89,041	21.7
Agricultural and industrial machinery mechanics	17,513	4.3
Machine tool regulators and operators	15,490	3.8
Welders and oxy-fuel cutting	14,491	3.5
Military police graduates	14,484	3.5
Couriers, baggage couriers	14,173	3.5
Military graduates	13,921	3.4
Gas station attendants	13,561	3.3
Auto bodyworker	11,500	2.8
Electrotechnicians	10,343	2.5
Welders and similar	9,982	2.4
Public transport inspectors and collectors	9,383	2.3
Total	409,812	100.0

(A) Race

Occupation	Count	Share(%)
Machine tool regulators and operators	15,490	6.5
Welders and oxy-fuel cutting	14,491	6.1
Couriers, baggage couriers	14,173	6.0
Gas station attendants	13,561	5.7
Auto bodyworkers	11,500	4.9
Joiners and the like	9,982	4.2
Public transport inspectors and collectors	9,383	4.0
Electrical workers and the like	8,193	3.5
Forklift operators	7,687	3.2
Land movement machine operators	7,449	3.1
Teachers' helpers	6,686	2.8
Polishers	6,516	2.8
Total	236,697	100.0

(B) Wage

Occupation	Count	Share(%)
General clerks	95,254	23.6
Elementary school teachers	74,423	18.4
Preschool teachers	25,938	6.4
High school teachers	20,819	5.1
Community health workers	19,407	4.8
Specialists in pedagogical methods	15,504	3.8
Executive and administrative secretaries	12,279	3.0
Nursing professionals	11,669	2.9
Mid-level legal professionals	8,176	2.0
Accounting and costing workers	7,458	1.8
University and higher education teachers	7,008	1.7
Beauty treatment specialists	6,896	1.7
Total	404,398	100.0

(C) Gender

Notes: The tables shows the distribution of top occupations in our alternative control groups, ranked by the number of times each occupation is included in each group. The data comes from the PNAD Continua rotating panel dataset from 2012 to 2016.

Table A4: Difference-in-differences effects of CA 72 and CL 150 on domestic workers

	Work Card	Soc. Security	Log Real Wage	Weekly Hours
Post CA/Pre CL	0.85*	1.48***	0.039***	-1.17***
	(0.49)	(0.44)	(0.0038)	(0.087)
Post CL	3.59***	4.79***	0.068***	-1.21***
	(0.49)	(0.43)	(0.0056)	(0.28)
Observations	571303	603376	594210	603376

	Works Overtime	Wage Equal to MW
Post CA/Pre CL	-2.31***	1.71***
	(0.32)	(0.60)
Post CL	-0.76	4.50***
	(0.90)	(0.92)
Observations	604339	246140

Notes: Standard errors clustered by occupation in parentheses. Stars denote statistical significance (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). The table shows difference-in-differences (DID) coefficients for the effects of CA 72 and CL 150. Two post periods are considered: "Post CA/Pre CL" (after CA 72 but before CL 150) and "Post CL" (after CL 150). The variables presented are an interaction with the treatment group variable; non-interacted versions are omitted. Covariates include individual-level variables: third degree polynomials of schooling and age, plus binary variables for female and white. In addition, it includes third-degree polynomials of the leave-out mean (at the neighborhood – or UPA – level) of years of schooling, age, real wages, female, and white. We include fixed effects for occupation, quarter, and neighborhood. Dependent variables are: Work Card (work permit definition of formality), Soc. Security (social security contribution definition of formality), Log Real Wage (log of real monthly wage), and Weekly Hours (hours worked per week). In addition, we include two binary variables, Works Overtime and Wage Equal to MW.

Table A5: Relationship of neighborhood peers with formalization measures

	Work Card	Soc. Security
DW	-49.0	-56.4
	(0.23)	(0.21)
Other DWs	-0.15	-0.56
	(0.057)	(0.053)
Other DWs × DW	2.18	2.37
	(0.072)	(0.068)
Formal Non-DWs	2.31	1.65
	(0.017)	(0.016)
Formal Non-DWs × DW	-0.21	1.07
	(0.025)	(0.024)
Observations	645797	677831

Note: These regressions include no controls. Standard errors in parentheses. Dependent variables are Work Card (work permit definition of formality) and Soc. Security (social security contribution definition of formality). The table displays the relationship between the number of other domestic workers (Other DWs) and formal non-domestic workers (Formal Non-DWs) in the neighborhood and formality measure. DW is a binary variable for being a domestic worker.

Table A6: Relationship of neighborhood formalization events with formalization measures

	Work Card	Soc. Security
DW	-50.2	-48.6
	(0.14)	(0.14)
Formal Events	1.74	1.04
	(0.20)	(0.19)
Formal Events × DW	0.58	1.57
	(0.29)	(0.28)
DW Formal Events	6.79	4.15
	(0.56)	(0.52)
DW Formal Events × DW	5.94	9.08
	(0.75)	(0.71)
Observations	533829	560452

Note: These regressions include no controls. Standard errors in parentheses. The dependent variables are Work Card (work permit definition of formality) and Soc. Security (social security contribution definition of formality). The table displays the relationship between having other domestic workers (DW Formal Events) and non-domestic workers (Formal Events) who became formal in the neighborhood in the past 2 quarters and formality measures. DW is an indicator for being a domestic worker.

Appendix B: Model Proofs

B.1 Proof of Theorem 1

Note that the effect of a change in any parameter ξ on the formality rate $s_F = \frac{P_F}{P_F+P_I}$ can be expressed as:

$$\frac{\partial s_F}{\partial \xi} = \frac{P_I}{(P_F+P_I)^2} \cdot \frac{\partial P_F}{\partial \xi} - \frac{P_F}{(P_F+P_I)^2} \cdot \frac{\partial P_I}{\partial \xi} \tag{11}$$

We can also express the effect of changing ξ on the choice probabilities using the multinomial logit properties (Train, 2009):

$$\frac{\partial P_F}{\partial \xi} = P_F \left[\frac{\partial V_F}{\partial \xi} - \sum_{k \in \{F,I,N\}} P_k \frac{\partial V_k}{\partial \xi} \right] \tag{12}$$

$$\frac{\partial P_I}{\partial \xi} = P_I \left[\frac{\partial V_I}{\partial \xi} - \sum_{k \in \{F,I,N\}} P_k \frac{\partial V_k}{\partial \xi} \right] \tag{13}$$

$$\frac{\partial P_N}{\partial \xi} = P_N \left[\frac{\partial V_N}{\partial \xi} - \sum_{k \in \{F,I,N\}} P_k \frac{\partial V_k}{\partial \xi} \right] \tag{14}$$

where $V_F = y - \bar{w}(1+\tau)$, $V_I = y - w_I - C_L = y - b_w - \theta(C_L)(\bar{w} - b_w) - C_L$, and $V_N = b_e$ represent the deterministic components of the utilities for formal employment, informal employment, and no job, respectively. The expected legal cost of informal employment is:

$$C_L(E) = H\left(\frac{pF}{1-E}\right) pF. \tag{15}$$

B.1.1 Effect of Taxes and Mandated Benefits (τ)

Taxes and mandated benefits τ appear only in the formal employment utility $V_F = y - \bar{w}(1+\tau)$, giving:

$$\frac{\partial V_F}{\partial \tau} = -\bar{w} \quad \text{and} \quad \frac{\partial V_I}{\partial \tau} = \frac{\partial V_N}{\partial \tau} = 0 \tag{16}$$

This yields:

$$\frac{\partial P_F}{\partial \tau} = -\bar{w} \cdot P_F \cdot (1 - P_F) < 0 \text{ and } \frac{\partial P_I}{\partial \tau} = \bar{w} \cdot P_F \cdot P_I > 0 \tag{17}$$

Substituting into our previous formula:

$$\frac{\partial s_F}{\partial \tau} = \frac{P_I}{(P_F + P_I)^2} \cdot (-\bar{w} \cdot P_F \cdot (1 - P_F)) - \frac{P_F}{(P_F + P_I)^2} \cdot (\bar{w} \cdot P_F \cdot P_I) \tag{18}$$

$$= -\frac{\bar{w} \cdot P_F \cdot P_I}{(P_F + P_I)^2} \cdot (1 - P_F + P_F) \tag{19}$$

$$= -\frac{\bar{w} \cdot P_F \cdot P_I}{(P_F + P_I)^2} < 0 \tag{20}$$

Thus, higher taxes and benefits decrease the formality rate. For the effect on unemployment, notice that:

$$\frac{\partial P_N}{\partial \tau} = \bar{w} \cdot P_F \cdot P_N > 0 \tag{21}$$

This shows that higher taxes and benefits increase nonemployment.

B.1.2 Effect of Worker Enforcement Capacity (E)

Worker enforcement capacity E affects only the informal employment utility, and it does so through the employer's expected legal cost C_L, which enters V_I both directly and through the informal wage $w_I = b_w + \theta(C_L)(\bar{w} - b_w)$. Since

$$\frac{\partial}{\partial E}\left(\frac{pF}{1-E}\right) = \frac{pF}{(1-E)^2} > 0, \tag{22}$$

we have

$$\frac{\partial C_L}{\partial E} = \frac{(pF)^2}{(1-E)^2} H'\left(\frac{pF}{1-E}\right) > 0. \tag{23}$$

Differentiating $V_I = y - b_w - \theta(C_L)(\bar{w} - b_w) - C_L$:

$$\frac{\partial V_F}{\partial E} = 0 \quad \text{and} \quad \frac{\partial V_I}{\partial E} = -\left[\theta'(C_L)(\bar{w} - b_w) + 1\right] \frac{\partial C_L}{\partial E} < 0 \quad \text{and} \quad \frac{\partial V_N}{\partial E} = 0. \tag{24}$$

where the sign follows from $\theta' > 0$, $\bar{w} > b_w$, and $\frac{\partial C_L}{\partial E} > 0$. Let:

$$\gamma_E = \left[\theta'(C_L)(\bar{w} - b_w) + 1\right] \frac{\partial C_L}{\partial E} > 0. \tag{25}$$

This yields:

$$\frac{\partial P_F}{\partial E} = \gamma_E \cdot P_F \cdot P_I > 0 \text{ and } \frac{\partial P_I}{\partial E} = -\gamma_E \cdot P_I \cdot (1 - P_I) < 0 \tag{26}$$

Substituting into our previous formula:

$$\frac{\partial s_F}{\partial E} = \frac{P_I}{(P_F + P_I)^2} \cdot (\gamma_E \cdot P_F \cdot P_I) - \frac{P_F}{(P_F + P_I)^2} \cdot (-\gamma_E \cdot P_I \cdot (1 - P_I)) \tag{27}$$

$$= \gamma_E \cdot \left[\frac{P_F \cdot P_I^2}{(P_F + P_I)^2} + \frac{P_F \cdot P_I \cdot (1 - P_I)}{(P_F + P_I)^2}\right] \tag{28}$$

$$= \gamma_E \cdot \frac{P_F \cdot P_I}{(P_F + P_I)^2} > 0 \tag{29}$$

Thus, greater worker enforcement capacity increases the formality rate. For the effect on unemployment:

$$\frac{\partial P_N}{\partial E} = \gamma_E \cdot P_I \cdot P_N > 0 \tag{30}$$

This shows that greater worker enforcement capacity increases nonemployment.

B.2 Proof of Theorem 2

We begin by establishing the framework for analyzing effects on the average wage. For any parameter ξ, and given that $\bar{w}$ is exogenous, the effect on the average wage $\bar{W} = s_F \cdot \bar{w} + (1 - s_F) \cdot w_I$ can be decomposed into direct and composition effects:

$$\frac{\partial \bar{W}}{\partial \xi} = \frac{\partial \bar{W}}{\partial w_I} \cdot \frac{\partial w_I}{\partial \xi} + \frac{\partial \bar{W}}{\partial s_F} \cdot \frac{\partial s_F}{\partial \xi} \tag{31}$$

where:

$$\frac{\partial \bar{W}}{\partial w_I} = (1 - s_F) = \frac{P_I}{P_F + P_I} \tag{32}$$

and

$$\frac{\partial \bar{W}}{\partial s_F} = \bar{w} - w_I. \tag{33}$$

B.2.1 Effect of Taxes and Mandated Benefits (τ)

Taxes and mandated benefits τ do not directly affect the informal wage:

$$\frac{\partial w_I}{\partial \tau} = 0. \tag{34}$$

From Theorem 1, we know that:

$$\begin{aligned}
\frac{\partial \bar{W}}{\partial \tau} &= 0 + (\bar{w} - w_I) \cdot \frac{\partial s_F}{\partial \tau} && (35)\\
&= (\bar{w} - w_I) \cdot \left(-\frac{\bar{w} \cdot P_F \cdot P_I}{(P_F + P_I)^2} \right) && (36)\\
&= -(\bar{w} - w_I) \cdot \frac{\bar{w} \cdot P_F \cdot P_I}{(P_F + P_I)^2}. && (37)
\end{aligned}$$

Given that $\bar{w} > w_I$, we have $\frac{\partial \bar{W}}{\partial \tau} < 0$. This proves that higher taxes and benefits decrease average wages.

B.2.2 Effect of Worker Enforcement Capacity (E)

Worker enforcement capacity raises the informal wage through its effect on the expected legal cost of informality:

$$\frac{\partial w_I}{\partial E} = \theta'(C_L)(\bar{w} - b_w)\frac{\partial C_L}{\partial E} > 0. \tag{38}$$

From Theorem 1, the effect on the average wage combines direct and compositional effects:

$$\begin{aligned}
\frac{\partial \bar{W}}{\partial E} &= \frac{P_I}{P_F + P_I} \cdot \theta'(C_L)(\bar{w} - b_w)\frac{\partial C_L}{\partial E} + (\bar{w} - w_I) \cdot \frac{\partial s_F}{\partial E} && (39)\\
&= \frac{P_I}{P_F + P_I} \cdot \theta'(C_L)(\bar{w} - b_w)\frac{\partial C_L}{\partial E} + (\bar{w} - w_I) \cdot \gamma_E \cdot \frac{P_F \cdot P_I}{(P_F + P_I)^2}. && (40)
\end{aligned}$$

The first term is positive. Given that $\bar{w} > w_I$, the second term is also positive. Therefore, $\frac{\partial \bar{W}}{\partial E} > 0$. This proves that greater worker enforcement capacity increases average wages.